

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams ON Acta Limnologica Brasiliensia

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Abstract

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1 Introduction

Essas abordagens são baseadas nos conceitos de turnover and nestedness. Species turnover and nestedness are two distinct components used to break down and understand beta diversity—which measures how much species composition changes between different communities or sites

(Baselga 2010), where, turnover is the replacement of some species by different species from one site to another, while nestedness is the loss or gain of species where poorer sites contain a strict subset of the species found at richer sites.

CITAR: (Gering et al. 2003, Crist et al. 2003, Veech et al. 2003, Ligeiro et al. 2010, Baselga 2010, 2012, Ávila et al. 2011, Shell and Collier 2018)

1.1 Objectives

This study describes the richness and density of zooplankton, comparatively between two small head water streams subject to different levels of environmental pressures. One originates within a protected area - Biological Reserve, or REBIO, according to Brasil (2000) - and its course extends beyond its administrative boundaries (MMA 2003), and a second one that flows entirely outside of the protected area. Within this framework, we use hierarchical additive partitioning to evaluate the relative contribution of different spatial and temporal scales to the diversity of zooplankton assemblages. Given the importance of the Atlantic Forest to the biodiversity (Pereira et al. 2026) and its fragmented stated north of the São Francisco river basin (Silva and Casteleti 2003), we aim with this study, (1) to infer environmental degradation from species turnover (Baselga 2010) and (2) to associate beta-diversity to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. Given the expected different levels of environmental pressures in the area (Gouveia et al. 2017), we hypothesize that the diversity is not be homogeneously distributed among the different spatial scales but concentrated at the finer levels and seasonal variation will be limited to species composition, but no spacial variation is expected between dry and wet seasons. Some environmental features (reach morphology, habitat structure, and water quality) are also described to explore possible explaining factors for the observed patterns.

2 Material and methods

2.1 Study area and sampling design

This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. This protected area is located in a transitional zone between the Caatinga and Atlantic Forest ecoregions. The land use of the study area is based on the occupation of traditional human populations and an increasing use of land for large scale monocultures, mostly sugar cane (Heinrichs et al. 2017). Other than these, small scale agricultural activities, fishing, and the modification of the natural vegetation for livestock are important anthropogenic pressures that affect the buffer zone of the REBIO (Arruda et al. 2013, MMA 2026). Other common uses of the soil in the area are, coconut plantations, orchards and subsistence agriculture. These land uses represent important sources of pollution, either through agrochemicals runoff or direct and indirect domestic effluents (such

as septic tanks). These nonpoint source pollutants contaminate underground water and create a spatial gradient of environmental disturbance (Mehla et al. 2026). Therefore, the mainly rural nature of the land use associated with the management of vegetation, will cause direct impact to aquatic systems, by the siltation of streams and contamination of groundwater. The hydrographic network of the study region is composed predominantly of small catchments containing first- and second-order streams (Strahler 1957 classification) (Figure 1). These watercourses are tributaries of the Camaratuba River and are sustained by numerous springs distributed throughout the region. The two main stream systems in the study area are the Barro Branco and Caiana catchments. The Barro Branco stream is approximately 8.8 km long and originates within the protected area. Its upper section remains within the boundaries of the REBIO, whereas its middle and lower sections extend into the surrounding landscape. Despite this transition, riparian vegetation is preserved along most of the stream. The Caiana stream is approximately 9.8 km long and also originates within the REBIO, although most of its course lies outside the protection area.

The climate in the region is classified as As' (Peel et al. 2007), characterized by a mean annual temperature of approximately 25 °C, with maximum temperatures occurring mainly between February and March, during the dry period, as well as by low annual thermal amplitude. Annual precipitation ranges from 1,400 to 1,800 mm, concentrated between March and July, whereas the lowest rainfall levels occur between September and December. In the study area, rainfall is concentrated between March and July, beginning in autumn and extending into winter. This pattern is attributed to the influence of cold air masses from the South Atlantic ocean that affect the region. The hydrographic network is composed of small headwater streams fed by rain (Gouveia et al. 2017).

A total of 15 sampling sites distributed longitudinally across the two Atlantic Forest streams were selected to capture spatial variation in environmental conditions and habitat characteristics. The Barro Branco stream included sampling sites located within a protected area (REBIO Guaribas) and sites located outside its boundaries, whereas all sampling sites in the Caiana stream were located outside the protected area. This sampling design allowed the assessment of zooplankton diversity across nested spatial scales and contrasting conditions of environmental protection and human influence (Figure 1). Sampling included temporal (dry and wet seasons) and spatial scales. The spatial scales of zooplankton diversity were represented by the total gamma () diversity, encompassing both study streams. At each stream, several sites were selected according to their position in the main channel of the catchment basin and, at each site, three habitats were sampled to represent different habitat types (pool, riffle, near underwater structures) (Frissell et al. 1986). We also considered a further spatial scale being the upper reaches (portion of the Barro Branco stream within the conservation unit and the Caiana stream before its confluence with the Pipina stream) and the lower reaches (portion of the Barro Branco stream outside the conservation unit and the Caiana stream after its confluence with the Pipina stream) (Figure 1). Sampling was conducted during the dry (January-February) and wet (June-July) seasons of 2024.

2.2 Environmental data collection

Environmental conditions were characterized using four groups of variables, following procedures adapted by Medeiros et al. (2008) from the protocols proposed by Pusey et al. (2004) and Mugodo et al. (2006): physicochemical water quality, stream site morphology, substrate composition, and marginal habitat structure. Water quality was characterized using physical and chemical measurements obtained with portable instruments, including water temperature ($^{\circ}\text{C}$) and dissolved oxygen concentration (mg/L). Turbidity was subsequently measured in the laboratory with a turbidimeter using water samples collected at each sampling site. Current velocity (m/s) was estimated using the float method, as described by Maitland (1990). Morphological characteristics were assessed from three randomly selected transects at each site, where stream width (cm) and water depth (cm) were measured. Bank slope was classified into four categories according to its inclination: 1, 20° ; 2, $> 20\text{--}30^{\circ}$; 3, $> 30\text{--}50^{\circ}$; and 4, $> 50^{\circ}$. Broader-scale landscape characteristics, including elevation and stream length, were obtained from handheld GPS records and satellite imagery. Substrate composition and the physical structure of marginal habitats were evaluated from 9 to 12 quadrats (1 m^2) distributed along the terrestrial-aquatic interface at each sampling site. Within each quadrat, the proportional cover of different substrate categories was visually estimated, including mud, sand, cobbles, fine and coarse gravel and rocks. The same quadrats were used to characterize littoral and submerged habitat features. Recorded habitat components included macrophytes, marginal grasses, leaf litter, attached and filamentous algae, overhanging and submerged vegetation, small and large woody debris, and root masses (Medeiros et al. 2008). For subsequent analyses and presentation, environmental measurements were summarized as mean values for each sampling site.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L^{-1}) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (1998) and Mackereth et al. (1978).

2.3 Zooplankton sampling

Zooplankton was sampled at habitats within each sampling site, hereafter referred to as habitat samples. Collections were performed in the main channel of the streams at the surface water. At each sampling habitat, a plankton net (30 cm mouth diameter, 70 cm length, and 60 μm mesh size) was used to filter 50 L of water. The net was rinsed between successive samples to prevent clogging and carryover of organisms. Each sample was concentrated to a final volume of 80 mL for laboratory analyses. Following collection, zooplankton samples were anesthetized with commercial sparkling water and subsequently preserved in 4% formalin (EPA 2020). Sucrose was added to the preserved sample to prevent female cladocerans from losing eggs and to minimize carapace distortion (Haney and Hall 1973). In the laboratory, samples had their zooplankton identified and counted using a Sedgewick–Rafter counting chamber. The analyses

focused on representatives of Rotifera, Cladocera, and Copepoda. Taxonomic identification was based on specialized literature and confirmed by specialists. To evaluate whether the sampling effort adequately represented the zooplankton assemblage, species accumulation curves were calculated (Melo et al. 2014, Lucena et al. 2015).

2.4 Statistical analyses

Environmental variables were screened for multicollinearity and to identify redundant descriptors. Variables representing similar environmental information were also combined when appropriate, thereby reducing redundancy in the dataset (Legendre and Legendre 1998).

```
1  #ZOOEBIO23_PART - ORGANIZANDO DADOS AMBIENTAIS----
2  #####....-----
3
4  dev.off() #apaga os graficos, se houver algum
5  rm(list=ls(all=TRUE)) #limpa a memória
6  cat("\014") #limpa o console
7
8  ##CARREGANDO MATRIZES BRUTAS-----
9
10 library(openxlsx)
11 habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
12                     rowNames = T,
13                     colNames = T,
14                     sheet = "ambiente",
15                     startRow = 2)
16
17 # List columns with at least one NA
18 colnames(habitat)[colSums(is.na(habitat)) > 0]
19 # Identify columns containing only NA values
20 all.na <- names(habitat)[colSums(!is.na(habitat)) == 0]
21 all.na <- c(all.na, "w.Temp_C", "w.DO_mg.L")
22 #all.na <- c(all.na, "n.Ammonia_NH3_ug.L", "n.Nitrite_NO2_ug.L",
23 #            "n.Nitrate_NO3_ug.L", "n.Orthophosphate_SRP_ug.L",
24 #            "n.Phosphorus_TP_ug.L")
25 # Remove those columns by name
26 m_hab_part <- habitat[, !names(habitat) %in% all.na, drop = FALSE]
27 # Removendo linhas por nome
28 rownames(m_hab_part)
29 del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3",
30             "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
31 del_rows
```

```

32 m_hab_part <- m_hab_part[!(row.names(m_hab_part) %in% c(del_rows)),]
33 m_hab_part
34 # List columns with at least one NA
35 colnames(m_hab_part)[colSums(is.na(m_hab_part)) > 0]
36 #rownames(m_hab_part)[is.na(m_hab_part$m.Width_m)]
37 # Remove columns with at least one NA
38 m_hab_part <- m_hab_part[, colSums(is.na(m_hab_part)) == 0, drop = FALSE]
39
40 ###REMOVENDO COLUNAS ZERADAS DE HABITAT----
41
42 sum <- colSums(m_hab_part)
43 sum
44 zero_sum <- names(which(colSums(m_hab_part) == 0))
45 zero_sum #nomes das colunas zeradas
46 m_part_cols <- m_hab_part[(colSums(m_hab_part) != 0)] #em != a exclamação inverte o sentido
47 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
48 zero_sum2 #nomes das colunas zeradas
49 sum<-colSums(m_part_cols)
50 sum
51
52 t_grps <- read.xlsx("dados/rebio23-habitat.xlsx",
53                   rowNames = T,
54                   colNames = T,
55                   sheet = "grupos",
56                   startRow = 2)
57
58 ###REMOVENDO LINHAS ZERADAS DE GRUPOS----
59
60 rownames(t_grps)
61 del_rows
62 t_grps <- t_grps[!(row.names(t_grps) %in% c(del_rows)),]
63 t_grps
64
65 ###SEPARANDO NUTRIENTES----
66
67 nutri <- grep("^n\\.\\.", colnames(habitat), value = TRUE)
68 nutri
69 nutrientes <- habitat[, nutri, drop = FALSE]
70 nutrientes <- nutrientes[, colSums(!is.na(nutrientes)) > 0, drop = FALSE]
71 nutrientes <- na.omit(nutrientes)
72
73 ###SALVANDO MATRIZES FINAIS----
74

```

```

75 write.table(m_part_cols, "res/m_hab_part.csv",
76             sep = ";", dec = ".", #"\t",
77             row.names = TRUE,
78             quote = TRUE,
79             append = FALSE)
80 write.table(nutrientes, "res/t_nutri.csv",
81             sep = ";", dec = ".", #"\t",
82             row.names = TRUE,
83             quote = TRUE,
84             append = FALSE)
85 write.table(t_grps, "res/t_grps.csv",
86             sep = ";", dec = ".", #"\t",
87             row.names = TRUE,
88             quote = TRUE,
89             append = FALSE)
90 t_grps <- read.csv("res/t_grps.csv",
91                  sep = ";", dec = ".",
92                  row.names = 1,
93                  header = TRUE,
94                  na.strings = NA)
95 t_nutri <- read.csv("res/t_nutri.csv",
96                   sep = ";", dec = ".",
97                   row.names = 1,
98                   header = TRUE,
99                   na.strings = NA)
100 m_hab <- read.csv("res/m_hab_part.csv",
101                  sep = ";", dec = ".",
102                  row.names = 1,
103                  header = TRUE,
104                  na.strings = NA)
105
106 #ZOOEBIO23_PART - ORGANIZANDO DADOS DA COMUNIDADE----
107 #####....----
108
109 dev.off() #apaga os graficos, se houver algum
110 rm(list=ls(all=TRUE)) #limpa a memória
111 cat("\014") #limpa o console
112
113 library(openxlsx)
114
115 ##CARREGANDO MATRIZES BRUTAS----
116
117 densidade <- read.xlsx("dados/rebio23-zoopl.xlsx",

```

```

118         rowNames = T,
119         colNames = T,
120         sheet = "densidade_hab_indL")
121 densidade[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
122
123 ##COLUNAS ZERADAS DE DENSIDADE----
124 sum <- colSums(densidade)
125 sum
126 zero_sum <- names(which(colSums(densidade) == 0))
127 zero_sum #nomes das espécies zeradas
128 m_part_cols <- densidade[(colSums(densidade) != 0)] #em != a exclamação inverte o sentido
129 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
130 zero_sum2 #nomes das espécies zeradas
131 sum<-colSums(m_part_cols)
132 sum
133
134 ###LINHAS ZERADAS DE DENSIDADE----
135 m_part2 <- as.data.frame(t(m_part_cols))
136 sum <- colSums(m_part2)
137 sum
138 zero_sum <- names(which(colSums(m_part2) == 0))
139 zero_sum #nomes das espécies zeradas
140 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
141 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
142 zero_sum2 #nomes das espécies zeradas
143 sum<-colSums(m_part_rows)
144 sum
145 m_dens <- as.data.frame((t(m_part_rows)))
146
147 ##SALVANDO MATRIZES FINAIS REMOVIDAS AS LINHAS/COLUNAS ZERADAS----
148
149 write.table(m_dens, "res/m_dens.csv",
150             sep = ";", dec = ".", #"\\t",
151             row.names = TRUE,
152             quote = TRUE,
153             append = FALSE)
154 m_dens <- read.csv(res/"m_dens.csv",
155                  sep = ";", dec = ".",
156                  row.names = 1,
157                  header = TRUE,
158                  na.strings = NA)
159
160 #ZOOREBIO23_PART - MATRIZ DE CONTAGEM----

```

```

161 #####....-----
162
163 library(openxlsx)
164 contagem <- read.xlsx("dados/rebio23-zoopl.xlsx",
165                       rowNames = T,
166                       colNames = T,
167                       sheet = "rebio23_80ml",
168                       startRow = 4)
169 contagem[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
170
171 ###COLUNAS ZERADAS DE CONTAGEM-----
172 sum <- colSums(contagem)
173 sum
174 zero_sum <- names(which(colSums(contagem) == 0))
175 zero_sum #nomes das espécies zeradas
176 m_part_cols <- contagem[(colSums(contagem) != 0)] #em != a exclamação inverte o sentido
177 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
178 zero_sum2 #nomes das espécies zeradas
179 sum<-colSums(m_part_cols)
180 sum
181
182 ###LINHAS ZERADAS DE CONTAGEM-----
183 m_part2 <- as.data.frame(t(m_part_cols))
184 sum <- colSums(m_part2)
185 sum
186 zero_sum <- names(which(colSums(m_part2) == 0))
187 zero_sum #nomes das espécies zeradas
188 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
189 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
190 zero_sum2 #nomes das espécies zeradas
191 sum<-colSums(m_part_rows)
192 sum
193 m_cont <- (t(m_part_rows))
194 m_cont
195 sum(m_cont)
196 ncol(m_cont)
197
198 ##SALVANDO MATRIZ FINAL DE CONTAGEM-----
199
200 write.table(m_cont, "res/m_cont.csv",
201             sep = ";", dec = ".", #"\\t",
202             row.names = TRUE,
203             quote = TRUE,

```

```

204         append = FALSE)
205 m_cont <- read.csv("res/m_cont.csv",
206                   sep = ";", dec = ".",
207                   row.names = 1,
208                   header = TRUE,
209                   na.strings = NA)
210 m_cont
211
212 #ZOOEBIO23_PART - ORGANIZANDO ESCALAS DA COMUNIDADE----
213 #####....----
214
215 dev.off() #apaga os graficos, se houver algum
216 rm(list=ls(all=TRUE)) #limpa a memória
217 cat("\014") #limpa o console
218
219 library(openxlsx)
220
221 ##CARREGANDO MATRIZES BRUTAS----
222
223 escalas <- read.xlsx("dados/rebio23-zoopl.xlsx",
224                     rowNames = T,
225                     colNames = T,
226                     sheet = "escalas_final",
227                     startRow = 2)
228
229 rownames(escalas)
230 del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3", "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
231 del_rows
232 escalas <- escalas[!(row.names(escalas) %in% c(del_rows)),]
233
234 write.table(escalas, "res/m_esc.csv",
235             sep = ";", dec = ".", #"\\t",
236             row.names = TRUE,
237             quote = TRUE,
238             append = FALSE)
239 m_esc <- read.csv("res/m_esc.csv",
240                  sep = ";", dec = ".",
241                  row.names = 1,
242                  header = TRUE,
243                  na.strings = NA)
244 m_esc
245
246 #REBIO23-part----

```

```

2
3  ##ORGANIZANDO DADOS-----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\014")
8
9  t_grps <- read.csv("res/t_grps.csv",
10                    sep = ";", dec = ".",
11                    row.names = 1,
12                    header = TRUE,
13                    na.strings = NA)
14  m_hab <- read.csv("res/m_hab_part.csv",
15                  sep = ";", dec = ".",
16                  row.names = 1,
17                  header = TRUE,
18                  na.strings = NA)
19
20  ##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS-----
21
22  library(psych)
23  colnames(m_hab)
24  #png("res/fig-h.hab_pairs.png")
25  pairs.panels(m_hab[,1:8],
26              method = "pearson", # correlation method
27              scale = FALSE, lm = FALSE,
28              hist.col = "#00AFBB", pch = 19,
29              density = TRUE, # show density plots
30              ellipses = TRUE, # show correlation ellipses
31              alpha = 0.5)
32  #dev.off()
33
34  cor <- cor(m_hab)
35  cor
36
37  library(corrplot)
38  #png("res/fig-hab_corrplot.png")
39  corrplot(cor, method = "circle")
40  #dev.off()
41
42  ##DELETANDO COLINEARES OU INDESEJADAS-----
43
44  #sink(file = "res/colineares.txt", append = F, split = T)

```

```

45 colnames(m_hab)
46 del_cols <- c("", "")
47 #del_cols <- grep("~n\\. ", colnames(m_hab), value = TRUE)
48 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
49
50 ##SOMANDO REDUNDANTES----
51
52 m_hab_part$h.Algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
53 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
54                               %in% c("h.Algae_f", "h.Algae_a"))]
55 m_hab_part$h.Debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s
56 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
57                               %in% c("h.Debris_l", "h.Debris_s"))]
58
59 colnames(m_hab_part)
60 m_hab_part
61 #sink()
62
63 write.table(m_hab_part, "res/m_hab_part.csv",
64             sep = ";", dec = ".", #"\\t",
65             row.names = TRUE,
66             quote = TRUE,
67             append = FALSE)
68 m_hab_part <- read.csv("res/m_hab_part.csv",
69                       sep = ";", dec = ".",
70                       row.names = 1,
71                       header = TRUE,
72                       na.strings = NA)
73
74 colnames(m_hab_part)
75 library(PerformanceAnalytics)
76 chart.Correlation(m_hab_part[, c(1:20)], histogram=TRUE)
77 # ***p<0.001, **p<0.01, *p<0.05, .p<0.10
78
79 library(Hmisc)
80 mat <- as.matrix(m_hab_part[, 1:20])
81 res <- rcorr(mat, type = "pearson")
82 cor_mat <- res$r
83 p_mat <- res$P
84
85 stars <- ifelse(p_mat < 0.001, "***",
86                 ifelse(p_mat < 0.01, "**",
87                         ifelse(p_mat < 0.05, "*", "")))

```

```

15
16 cor_star <- ifelse(p_mat < 0.05,
17                   paste0(round(cor_mat, 2), stars),
18                   NA)
19
20 cor_star <- matrix(
21   cor_star,
22   nrow = nrow(cor_mat),
23   dimnames = dimnames(cor_mat)
24 )
25
26 cor_star[upper.tri(cor_star, diag = TRUE)] <- NA
27 cor_star <- noquote(cor_star)
28 print(cor_star, na.print = "")

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_part <- read.csv("res/m_hab_part.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20
21 colnames(m_hab_part)
22
23 ###RELATIVIZAÇÕES E TRANSFORMAÇÕES----
24 library(vegan)
25 library(tidyverse)
26
27 # Selecionando variáveis que começam com "w" ou "m"
28 m_wm <- m_hab_part[, grep("^(w|m)", colnames(m_hab_part))]
```

```

29
30 # Selecionando variáveis que começam com "s" ou "h"
31 m_sh <- m_hab_part[, grep("^(s|h)", colnames(m_hab_part))]
32
33 # Removendo os dois primeiros caracteres dos nomes das colunas
34 colnames(m_wm) <- sub("^..", "", colnames(m_wm))
35 colnames(m_sh) <- sub("^..", "", colnames(m_sh))
36
37 # Transformações
38 m_wm <- sqrt(m_wm)
39 m_sh <- asin(sqrt(decostand(m_sh, method = "total", MARGIN = 2)))
40
41 # Juntando tudo em uma única tabela
42 hab_trns <- cbind(m_sh, m_wm)
43
44 # Salvando m_hab_trns
45 write.table(hab_trns, "res/m_hab_trns.csv",
46             sep = ";", dec = ".", #"\t",
47             row.names = TRUE,
48             quote = TRUE,
49             append = FALSE)
50 m_hab_trns <- read.table("res/m_hab_trns.csv",
51                          sep = ";", dec = ".",
52                          row.names = 1,
53                          header = TRUE,
54                          na.strings = NA)

```

These data were then analyzed using Principal Component Analysis (PCA) to summarize multivariate patterns of environmental variation among sampling sites. Prior to PCA, site morphology and water-quality variables were square-root transformed (Sokal and Rohlf 1995). Variables describing sediment composition and marginal habitat structure, expressed as percentages, were relativized by their column totals and subsequently subjected to the arcsine square-root transformation (McCune and Grace 2002). PCA was conducted in R using the **FactoMineR** package (Lê et al. 2008), with all variables centered and standardized to unit variance before analysis. The water quality parameters temperature and dissolved oxygen, and all nutrients were used to characterize differences between streams but were not used in the PCA because they were not measured for both seasons. A Permutational Multivariate Analysis of Variance (PERMANOVA, 999 permutations) was performed to test whether the overall set of environmental variables measured differed across the seasons and streams (Anderson and Walsh 2013, Freiry et al. 2020).

To compare the diversity of zooplanktonic taxa between streams, the Chao1 estimated richness ($\pm$ SE) and the curves of rarefied and extrapolated richness were generated (95% upper and

lower confidence intervals) using individual-based rarefaction curves (Gotelli and Colwell 2001, Freiry et al. 2020).

Differences in assemblage composition across the spatial hierarchy and seasons were explored using the Unweighted Pair Group Method with Averages (UPGMA) (Belbin and McDonald 1993, Diniz-Reis et al. 2022), based on Bray-Curtis dissimilarities calculated from the zooplankton density matrix (Ávila et al. 2011). Prior to the analyses, density data were relativized and square-root transformed to reduce the influence of highly abundant taxa and improve the distributional properties of the dataset (Rocha et al. 2012). To focus on spatial variation across sampling sites, habitat density data were pooled for this analysis. To assess how well the UPGMA dendrograms represented the original Bray-Curtis dissimilarities among sampling sites, the Cophenetic Correlation Coefficient (CC) was calculated (Sokal and Rohlf 1962, Sneath and Sokal 1973, Saraçlı et al. 2013), where a cophenetic matrix was constructed from each seasonal dendrogram using the *cophenetic* function in the **vegan** package in R. The cophenetic distance matrix was then correlated with the corresponding original Bray-Curtis dissimilarity matrix using Pearson’s product-moment correlation coefficient (Sokal and Rohlf 1962, Farris 1969, Silva and Dias 2013). To evaluate the strength of the hierarchical clustering structure obtained from the UPGMA dendrograms, the Agglomerative Coefficient (AC) was calculated for each season using the *coef.hclust* function from the **cluster** package in R (Kaufman and Rousseeuw 1990, Maechler et al. 2026). The AC ranges from 0 to 1, where values closer to 1 indicate a strong, well-defined clustering structure and values near 0 reflect weak or indistinct grouping.

Additive diversity partitioning was used to quantify the contribution of within- and between-unit diversity to total zooplankton diversity. In this framework, alpha diversity (α) represents diversity occurring within sampling units, whereas beta diversity (β) represents the contribution of differences among units at successive levels of the spatial hierarchy (Lande 1996, Crist et al. 2003). The hierarchical structure included three spatial scales. The finest scales consisted of the habitat samples within each site ($n = 3$), and of each site ($n = 6$, for the Barro Branco stream, and $n = 9$, for the Caiana stream). The intermediate scales corresponded to the upper and lower stream sections ($n = 2$), whereas the broadest scale represented differences among the two streams. Accordingly, the total diversity (γ) was decomposed into within-habitat diversity (α_1) and the successive beta components associated with differences among habitats (β_1), among sites within upper and lower stream reaches (β_2), and among streams (β_3). The contribution of each component was expressed as a percentage of total diversity (Crist et al. 2003, Farias et al. 2020). To investigate seasonal variation in the hierarchical structure of diversity, additive partitioning was performed separately for each season and for each stream (Melo and Medeiros 2013, Farias et al. 2020).

The overall diversity was taken to be the sum of the relative contributions of each scale in the hierarchical sampling design. Analysis of significance of the contribution of each hierarchical level was performed through individual-based randomizations (type I null model) (see Freiry et al. 2020, Melo and Medeiros 2013). The randomization process was repeated 999 times to obtain null distributions of the α and β diversity at each hierarchical level, and

used to test whether the observed values were higher or lower than expected by chance (Crist et al. 2003). The probabilities obtained from randomization tests were interpreted as P -values, as a low P -value ($\text{Prop}_{exp} > \text{Prop}_{obs} < 0.05$) indicates an observed diversity significantly different than that expected under the null model. On the other hand, a high P -value ($\text{Prop}_{exp} > \text{Prop}_{obs} > 0.95$) indicates an observed diversity not different than that expected by chance (Veech et al. 2002, Gering et al. 2003, Crist et al. 2003, Freiry et al. 2020). The results of the analysis were subject to a Monte Carlo permutation test (999 permutations) for validation. The additive partitioning of the diversity was carried out with the function *adipart* from package *vegan* (Oksanen et al. n.d.). All analyses were conducted in the statistical environment R (R Core Team 2017).

3 Results

3.1 Environmental variables and nutrient concentration

Water velocity ranged from 0 to 0.41 m/s, with the highest average recorded at site 14. Waters were, on average, well oxygenated (5.3 to 8.1 mg/L), and cool, with average temperatures between from 25.2 to 32.5 °C, the average being 27.7 °C. Conductivity ranged from 98.8 and 223.0 S/cm, pH was mostly neutral to slightly acidic (from 5.1 to 6.9) and turbidity was low (4.9 to 79.2 NTU). Regarding the morphological variables, elevation ranged from 41.5 to 112.8 m a.s.l., while the stream margins slope reached 50° for some sites. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs in the stream channel. Stream site width varied from 1 m across to 19.5 m in larger reservoirs. Overall, depths ranged from approx. 10-20 cm to a maximum of 1.55 m, on average. Substrate composition across sites consisted mainly of mud (reaching 100% at sites 1 and 2, in the Barro Branco stream) and sand being more common at the Caiana Stream (reaching 98.3% cover at site 14) (Table 1). Marginal and underwater habitat structures were diverse with most variables being common at the Barro Branco stream, namely aquatic macrophytes, filamentous and attached algae and littoral grass. Submerged and overhanging vegetation, leaf litter, root masses and wood debris were present at all sites (see Table 1).

```

1 #TABELA DE HABITAT----
2
3 dev.off()
4 rm(list=ls(all=TRUE))
5 cat("\014")
6
7 habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
8                     rowNames = T,
9                     colNames = T,
```

```

10         sheet = "ambiente",
11         startRow = 2)
12 m_hab_part <- read.csv("res/m_hab_part.csv",
13                       sep = ";", dec = ".",
14                       row.names = 1,
15                       header = TRUE,
16                       na.strings = NA)
17 t_grps <- read.csv("res/t_grps.csv",
18                  sep = ";", dec = ".",
19                  row.names = 1,
20                  header = TRUE,
21                  na.strings = NA)
22
23 m_hab_part$w.Temp_C <- habitat[rownames(m_hab_part), "w.Temp_C"]
24 m_hab_part$w.DO_mg.L <- habitat[rownames(m_hab_part), "w.DO_mg.L"]
25
26 library(dplyr)
27 library(tidyr)
28 m_trab <- m_hab_part %>%
29   rename_with(~ gsub("_", ".", .))
30 m <- m_trab %>%
31   group_by(PontoN = t_grps$PontoN) %>%
32   summarise(across(where(is.numeric),
33                     list(mean = ~ mean(.x, na.rm = TRUE), #ajustes celulas vazias
34                           min = ~ min(.x, na.rm = TRUE),
35                           max = ~ max(.x, na.rm = TRUE))),
36             .groups = 'drop') %>%
37   pivot_longer(
38     cols = -c(PontoN),
39     names_to = c("Variable", ".value"),
40     names_sep = "_"
41   )
42
43 m <- as.data.frame(m)
44 m_wide <- m %>%
45   mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),
46                                paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3)
47                                paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
48   unite("Location", PontoN, sep = "_") %>%
49   select(Variable, Location, stat_string) %>%
50   pivot_wider(names_from = Location, values_from = stat_string)
51
52 m_wide

```

```

53 m_wide <- as.data.frame(m_wide)
54 cols <- colnames(m_wide)
55 col_order <- c(
56   "Variable",
57   cols[cols != "Variable"][
58     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))]
59 ]
60 )
61 m_wide <- m_wide[, c(col_order)] #sequência das colunas
62 row_order <- c(
63   "w.Temp.C",
64   "w.DO.mg.L",
65   "w.Vel.m.s",
66   "w.Cond.uS.cm",
67   "w.Turb.ntu",
68   "w.pH",
69   "m.Elev.m",
70   "m.Slope",
71   "m.Depth.avg.cm",
72   "m.Depth.mar.cm",
73   "m.Depth.max.cm",
74   "m.Width.m",
75   "s.Mud",
76   "s.Sand",
77   "s.Gravel.f",
78   "s.Rocks",
79   "h.Macrophytes",
80   "h.Algae",
81   "h.Subm.veg",
82   "h.Overh.veg",
83   "h.Grass",
84   "h.Leaf.l",
85   "h.Root.masses",
86   "h.Debris"
87 )
88 m_wide <- m_wide[match(row_order, m_wide$Variable), ]
89 m_wide
90
91 # Salvado m_wide
92 write.table(m_wide, "res/m_wide_hab_to.txt",
93             sep = ";", dec = ".", #"\\t",
94             row.names = TRUE,
95             quote = TRUE,

```

```

96         append = FALSE)
97 m_wide_hab_to <- read.table("res/m_wide_hab_to.txt",
98                             sep = ";", dec = ".",
99                             row.names = 1,
100                             header = TRUE,
101                             na.strings = NA)
102
103 # Exportando dados para Excel
104 library(openxlsx)
105 #write.xlsx(m_wide, file = "res/tabela de habitat.xlsx", rowNames = FALSE)
106 wb <- loadWorkbook("res/tabela de habitat.xlsx")
107 writeData(wb, sheet = "Sheet 1", x = m_wide)
108 saveWorkbook(wb, "res/tabela de habitat.xlsx", overwrite = TRUE)
109
110 # Tabela final ajustada no Excel
111 library(readxl)
112 hab_gtttable <- read_excel(
113   path = "res/tabela de habitat.xlsx",
114   sheet = "tabela_final")
115 #range = "A1:P29"#ajustar para o tamanho da tabela final
116 hab_gtttable
117 library(gt)
118 hab_gtttable <- gt(hab_gtttable)
119 hab_gtttable <- sub_missing(hab_gtttable,
120                             columns = everything(), missing_text = "")
121 #hab_gtttable <- fmt_number(hab_gtttable, columns = "F.O.(%)", decimals = 1)
122 hab_gtttable
123 gtsave(hab_gtttable, "res/gt-hab_gtttable.xlsx.html")
124 saveRDS(hab_gtttable, "res/gt-hab_gtttable.xlsx.rds")

```



```

1 #TABELA DE HABITAT----
2 ##SEGUE DO SCRIPT ANTERIOR
3
4 # Escolher sumário de uma variavel
5 m
6 var <- "w.Vel.m.s"
7 m[m$Variable == var, "mean"] #cada valor de var
8 summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
9
10 # Escolher sumário de um grupo de variáveis do df m
11 vars <- unique(grep("^h\\. ", m$Variable, value = TRUE))
12 summaries <- list() #cria uma lista vazia para guardar os sumários
13 # Loop para cada variável do grupo e guarda em summaries

```

```

14 for (var in vars) {
15   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
16 }
17 #var is a temporary variable used in the for loop to iterate through
18 #each variable name that starts with "h."
19
20 summaries
21 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
22 colnames(summary_table) <- c("Min", "Q1", "Median", "Mean", "Q3", "Max")
23 #sink(file = "res/summary_w.txt", split = TRUE)
24 round(summary_table, 2)
25 #round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
26 #sink()
27
28 #Summario para as médias
29
30 dev.off()
31 rm(list=ls(all=TRUE))
32 cat("\014")
33
34 m_hab_part <- read.csv("res/m_hab_part.csv",
35                       sep = ";", dec = ".",
36                       row.names = 1,
37                       header = TRUE,
38                       na.strings = NA)
39 t_grps <- read.csv("res/t_grps.csv",
40                   sep = ";", dec = ".",
41                   row.names = 1,
42                   header = TRUE,
43                   na.strings = NA)
44
45 library(dplyr)
46 library(tidyr)
47
48 m_trab <- m_hab_part %>%
49   rename_with(~ gsub("_", ".", .))
50
51 # Compute averages per group
52 m_s <- m_trab %>%
53   group_by(PontoN = t_grps$PontoN) %>%
54   summarise(across(where(is.numeric), ~ mean(.x, na.rm = TRUE)),
55             .groups = 'drop') %>%
56   pivot_longer(

```

```

57     cols = -PontoN,
58     names_to = "Variable",
59     values_to = "mean"
60 )
61
62 # Format rounding and reshape to wide format
63 m_wide_s <- m_s %>%
64   mutate(
65     stat_string = ifelse(
66       Variable == "m.Vel.m.s",
67       as.character(round(mean, 3)),
68       as.character(round(mean, 1))
69     )
70   ) %>%
71   unite("Location", PontoN, sep = "_") %>%
72   select(Variable, Location, stat_string) %>%
73   pivot_wider(names_from = Location, values_from = stat_string)
74
75 m_wide_s <- as.data.frame(m_wide_s)
76
77 # Order columns numerically by Ponto number
78 cols <- colnames(m_wide_s)
79 col_order <- c(
80   "Variable",
81   cols[cols != "Variable"][
82     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
83   ]
84 )
85 m_wide_s <- m_wide_s[, col_order]
86
87 # Order rows according to pre-defined variable list
88 row_order <- c(
89   "w.Cond.uS.cm",
90   "w.Turb.ntu",
91   "w.pH",
92   "m.Elev.m",
93   "m.Vel.m.s",
94   "m.Slope",
95   "m.Depth.avg.cm",
96   "m.Depth.mar.cm",
97   "s.Mud",
98   "s.Sand",
99   "s.Gravel.f",

```

```

100   "s.Rocks",
101   "h.Macrophytes",
102   "h.Algae",
103   "h.Subm.veg",
104   "h.Overh.veg",
105   "h.Grass",
106   "h.Leaf.l",
107   "h.Root.masses",
108   "h.Debris"
109 )
110
111 m_wide_s <- m_wide_s[match(row_order, m_wide_s$Variable), ]
112
113 # View final table
114 m_wide_s

```

Principal Component Analysis (PCA) was used to summarize environmental variation among sampling sites and to identify variables contributing to the differences between streams (Figure 2) and seasons (Figure 3). The first two principal components explained 38.6% of the total variance, with the first axis accounting for 23.2%. In relation to the environmental variables of water quality, the Barro Branco stream was more associated with higher elevations, the predominance of muddy substrate and a narrower array of underwater structures, mostly algae, macrophytes, leaf litter and woody debris. The Caiana stream showed a wider range of variables affecting its waters, being associated with higher pH, turbidity and depths, as well as a predominance of sandy substrates and, to a lesser degree, root masses. Water velocity was also important for this stream. PERMANOVA showed significant differences in the environmental variables between the two study streams ($F_{1,88} = 7.8$, $R^2 = 0.08$, $p = 0.001$). Dry and wet seasons did not show clear segregation in the PCA, even though PERMANOVA indicated significance of differences between seasons ($F_{1,88} = 3.8$, $R^2 = 0.04$, $p = 0.008$).

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,

```

```

14         na.strings = NA)
15 m_hab_trns <- read.csv("res/m_hab_trns.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20
21 #PCA----
22
23 library("FactoMineR")
24 library("factoextra")
25
26 pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
27 print(pca)
28
29 eig.val <- get_eigenvalue(pca)
30 eig.val
31
32 #sink(file = "res/cor_matrix.txt", append = FALSE)
33 pca$var$cor
34 #sink()
35
36 fviz_eig(pca, addlabels = TRUE, ylim = c(0,30))
37
38 var <- get_pca_var(pca)
39 var
40
41 fviz_pca_var(pca, col.var = "black")
42
43 library("corrplot")
44 corrplot(var$cos2, is.corr=FALSE)
45
46 fviz_cos2(pca, choice = "var", axes = 1:2)
47
48 fviz_pca_var(pca, col.var = "cos2",
49             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
50             repel = TRUE # Avoid text overlapping
51 )
52
53 fviz_pca_var(pca, alpha.var = "cos2")
54
55 corrplot(var$contrib, is.corr=FALSE)
56

```

```

57 fviz_contrib(pca, choice = "var", axes = 1, top = 10)
58 fviz_contrib(pca, choice = "var", axes = 2, top = 10)
59 fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
60
61 fviz_pca_var(pca, col.var = "contrib",
62               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
63 )
64 fviz_pca_var(pca, alpha.var = "contrib")
65
66 ##GROUPING BY KMEANS----
67
68 set.seed(123)
69 res.km <- kmeans(var$coord, centers = 3, nstart = 25)
70 grp <- as.factor(res.km$cluster)
71 # Color variables by groups
72 fviz_pca_var(pca, col.var = grp,
73               palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
74               legend.title = "Cluster")
75
76 ind <- get_pca_ind(pca)
77 ind
78 ind$contrib
79
80 ##BIPLOTS----
81
82 fviz_pca_ind(pca)
83
84 fviz_pca_ind(pca, col.ind = "cos2",
85               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
86               repel = TRUE #avoid text overlapping (slow if many points)
87 )
88
89 fviz_pca_ind(pca, pointsize = "cos2",
90               pointshape = 21, fill = "#E7B800",
91               repel = TRUE #avoid text overlapping (slow if many points)
92 )
93
94 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
95               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
96               repel = TRUE #avoid text overlapping (slow if many points)
97 )
98
99 fviz_cos2(pca, choice = "ind")

```

```

100
101 fviz_contrib(pca, choice = "ind", axes = 1:2)
102
103 # Create a random continuous variable of length 23,
104 # Same length as the number of active individuals in the PCA
105 t_grps <- t_grps[rownames(pca$ind$coord), ]
106 colnames(t_grps)
107 set.seed(123)
108 # Escolhendo variável categorica
109 my.cont.var <- t_grps$Periodo
110 my.cont.var[my.cont.var == "seco"] <- "dry"
111 my.cont.var[my.cont.var == "chuvoso"] <- "wet"
112 # Color individuals by the continuous variable
113 fviz_pca_ind(
114   pca,
115   habillage = my.cont.var,
116   palette = c("blue", "yellow", "red", "pink"),
117   repel = FALSE
118 )
119
120 fviz_pca_ind(pca,
121   geom.ind = "point", #show points only (nbut not "text")
122   col.ind = as.factor(my.cont.var), #color by groups
123   palette = "grey",
124   addEllipses = TRUE, #concentration ellipses
125   legend.title = "Groups"
126 )
127
128 length(unique(my.cont.var))
129 fviz_pca_ind(pca,
130   geom.ind = "point",
131   col.ind = my.cont.var,
132   palette = "grey",
133   addEllipses = TRUE, ellipse.type = "confidence",
134   mean.point = TRUE,
135   legend.title = "Groups")
136
137 fviz_pca_biplot(pca, repel = TRUE,
138   col.var = "#2E9FDF", #variables color
139   col.ind = "#696969" #individuals color
140 )
141
142 # PCA com ellipse colorida

```

```

143 fviz_pca_biplot(pca,
144                 col.ind = my.cont.var, palette = "jco",
145                 addEllipses = TRUE, ellipse.type = "t", #t-distribution confidence ellipses
146                 ellipse.level = 0.95,
147                 shape.ind = my.cont.var,
148                 label = "var",
149                 col.var = "black", repel = TRUE,
150                 legend.title = "Stream") +
151 ggtitle(NULL) +
152 scale_color_manual(values = c(
153   "#1B9E77",
154   "#D95F02",
155   "#7570B3",
156   "#E7298A"
157 )) +
158 scale_shape_manual(values = c(
159   16, # circle
160   17, # triangle
161   15, # square
162   18 # diamond
163 ))
164
165 ### FILTRANDO AS VARIÁVEIS
166 var_coord <- pca$var$coord[,1:2]
167 keep_vars <- apply(abs(var_coord) > 0.2,1,any)
168 vars_to_keep <- rownames(var_coord)[keep_vars]
169 pca_filt <- pca
170 pca_filt$var$coord <- pca$var$coord[vars_to_keep,,drop=FALSE]
171 pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep,,drop=FALSE]
172 pca_filt$var$contrib <- pca$var$contrib[vars_to_keep,,drop=FALSE]
173
174 ### AJUSTE MANUAL DAS COORDENADAS DOS VETORES
175
176 # Cria uma tabela SEPARADA apenas para as posições dos rótulos
177 label_coords <- as.data.frame(pca_filt$var$coord)
178 # Multiplica APENAS a posição do texto (as setas continuam intactas no pca_filt)
179 label_coords <- label_coords * 2.0
180 label_coords <- label_coords * 2.0
181 # Se quiser ajustar pontos específicos manualmente
182 #fix(label_coords) #nomes das colunas
183 #write.table(label_coords,"res/coords_lab.txt")
184 label_coords <- read.table("res/coords_lab.txt",
185                          header=TRUE,

```

```

186         row.names=1)
187
188 # GRÁFICO FINAL
189
190 #png("res/fig-hab_PCA_perodo.png", width = 9, height = 7.5, units = "in", res = 300)
191 fviz_pca_biplot(pca_filt,
192   col.ind = my.cont.var, palette = "jco",
193   addEllipses = TRUE, ellipse.type = "t",
194   ellipse.level = 0.95,
195   shape.ind = my.cont.var,
196   label = "none",
197   col.var = "black",
198   legend.title = "Season"
199 ) +
200   # Adiciona os rótulos nas posições customizadas da tabela label_coords
201   geom_text(
202     data = label_coords,
203     aes(x = Dim.1, y = Dim.2, label = rownames(label_coords)),
204     color = "black",
205     size = 12) +
206   ggtitle(NULL) +
207   scale_color_manual(values = c("#1B9E77", "#D95F02", "#7570B3", "#E7298A")) +
208   scale_fill_manual(values = c("darkcyan", "orange", "#7570B3", "#E7298A")) +
209   scale_shape_manual(values = c(
210     16, # circle
211     17, # triangle
212     15, # square
213     18 # diamond
214   )) +
215   theme(
216     axis.title = element_text(size = 34),
217     axis.text = element_text(size = 32),
218     legend.title = element_text(size = 40),
219     legend.text = element_text(size = 40)
220   )
221   #dev.off()
222
223 1 #ORGANIZANDO DADOS----
224 2
225 3 dev.off()
226 4 rm(list=ls(all=TRUE))
227 5 cat("\014")
228 6

```

```

7 t_grps <- read.csv("res/t_grps.csv",
8                     sep = ";", dec = ".",
9                     row.names = 1,
10                    header = TRUE,
11                    na.strings = NA)
12 m_hab_trns <- read.csv("res/m_hab_trns.csv",
13                       sep = ";", dec = ".",
14                       row.names = 1,
15                       header = TRUE,
16                       na.strings = NA)
17
18 colnames(t_grps)
19 m_trab <- m_hab_trns
20 library(vegan)
21
22 #MRPP----
23
24 #TRABALHANDO COM MULTIPLAS VARIAVEIS----
25
26 # Run MRPP for multiple variables
27 fatores <- c("Periodo", "Rio", "Ref_site")
28 set.seed(123)
29 mrpp_results <- lapply(fatores, function(v) {
30   grp <- as.factor(t_grps[[v]])
31   mrpp(
32     dat = m_trab,
33     grouping = grp,
34     distance = "bray"
35   )
36 })
37 names(mrpp_results) <- fatores
38 mrpp_results
39 # Print one result
40 mrpp_results$Ref_site
41 # Loop through summaries
42 sink(file = "res/mrpp.txt", split = TRUE)#, append = TRUE)
43 for (v in names(mrpp_results)) {
44   cat("\n===== \n")
45   cat("MRPP for:", v, "\n")
46   print(mrpp_results[[v]])
47 }
48 sink()
49 # Extract key statistics into a table

```

```

50 mrpp_summary <- data.frame(
51   Variable = fatores,
52   Delta = sapply(mrpp_results, function(x) x$delta),
53   Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
54   A = sapply(mrpp_results, function(x) x$A),
55   P_value = sapply(mrpp_results, function(x) x$Pvalue)
56 )
57 sink("res/mrpp.txt", append = TRUE)
58 mrpp_summary
59 sink()
60
61 ##TRABALHANDO COM UMA UNICA VARIABEL----
62
63 # Run MRPP for a single variable
64 grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
65 set.seed(123)
66 mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
67 mrpp
68 summary(mrpp)
69
70 # Teste de hipótese para uma unica variavel
71 str(mrpp)
72 p_value <- mrpp$Pvalue
73 print(sprintf("p-value: %.10f", p_value))
74 # Conditional statement to check the p-value and print the appropriate message
75 if (p_value < 0.05) {
76   print(sprintf("Existe diferença significativa porque o valor de p é %.10f, sendo menor que
77 } else {
78   print(sprintf("Não existe diferença significativa porque o valor de p é %.10f, sendo maior
79 }
80
81 ##TESTES PAREADOS----
82
83 # Todos os pares de níveis do fator
84 pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
85 pairs
86 # Aplicar mrpp para cada par
87 res_list <- lapply(pairs, function(p) {
88   sel <- grp %in% p
89   mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
90 })
91 # Dar nomes aos resultados
92 names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))

```

```

93  # visualizar
94  res_list
95  # Resumo
96  summary_tab <- data.frame(
97    Comparison = names(res_list),
98    A = sapply(res_list, function(x) x$A),
99    Delta = sapply(res_list, function(x) x$delta),
100    Pvalue = sapply(res_list, function(x) x$Pvalue)
101  )
102  sink("res/mrpp.txt", append = TRUE)
103  summary_tab
104  sink()
105
106  ###MATRIX-STYLE TABLE----
107  library(dplyr)
108  library(tidyr)
109  library(knitr)
110  library(kableExtra)
111  # Extract unique group names from the comparisons
112  groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
113  # Initialize empty matrix
114  pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
115                        dimnames = list(groups1, groups1))
116  # Fill matrix with Pvalues
117  for (i in 1:nrow(summary_tab)) {
118    comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
119    g1 <- comp[1]
120    g2 <- comp[2]
121    p <- summary_tab$Pvalue[i]
122
123    pval_matrix[g1, g2] <- p
124    pval_matrix[g2, g1] <- p #make symmetric
125  }
126
127  # Format p-values: red if <0.05
128  pval_colored <- ifelse(is.na(pval_matrix), "",
129                        ifelse(pval_matrix < 0.05,
130                              paste0("<span style='color:red;'>", sprintf("%.3f", pval_matrix)),
131                              sprintf("%.3f", pval_matrix)))
132
133  # Convert to data.frame for kable
134  pval_df <- as.data.frame(pval_colored)
135  pval_df <- tibble::rownames_to_column(pval_df, "Group")

```

```

136 # Render HTML table
137 matrixstyle <- pval_df %>%
138   kable("html", escape = FALSE) %>%
139   kable_styling(full_width = FALSE)
140 matrixstyle
141 save_kable(matrixstyle, "res/kb-mrpp_matrixstyle.html")
142
143 ##GRÁFICO DA MRPP----
144
145 png("res/fig-mrpp.png", width = 15, height = 10, units = "in", res = 300)
146 def.par <- par(no.readonly = TRUE)
147 layout(matrix(1:2,nr=1))
148 plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
149 with(t_grps, ordihull(ord, grp))
150 with(mrpp, {
151   fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
152                   main="Test of Differences Among Groups")
153   abline(v=delta);
154   text(delta, 2*mean(fig.dist$counts), adj = -0.5,
155         expression(bold(delta)), cex=1.5 ) }
156 )
157 par(def.par)
158 dev.off()
159
160 ###AVALIANDO AS DISTÂNCIAS MÉDIAS----
161 md <- with(t_grps, meandist(vegdist(m_trab), grp))
162 md
163 summary(md)
164 par(mfrow=c(1,2))
165 plot(md)
166 plot(md, kind="histogram")
167 par(mfrow=c(1,1))
168
169 #PerMANOVA----
170
171 #browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
172 #browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")
173
174 library(tidyverse)
175 library(ggtext)
176 library(ggplot2)
177 library(vegan)
178

```

```

179 adonis2(m_trab ~ Periodo, data = t_grps, method = "bray")
180 set.seed(123)
181 perm <- 999
182
183 all_test <- adonis2(m_trab ~ Periodo, data = t_grps, method = "bray", permutations = perm)
184 all_test
185
186 ## TESTES PARA VARIÁVEIS COM DOIS FATORES
187
188 sink(file = "res/perma.txt", split = TRUE)#, append = TRUE)
189 set.seed(123)
190 perm <- 999
191 adonis2(m_trab ~ Periodo, data = t_grps, method = "bray", permutations = perm)
192 adonis2(m_trab ~ Rio, data = t_grps, method = "bray", permutations = perm)
193 adonis2(m_trab ~ Periodo*Rio, data = t_grps)
194 adonis2(m_trab ~ Periodo*Rio, data = t_grps, by = "terms") #ou by=NULL remove as interações
195
196 ## AVALIANDO INTERAÇÕES-----
197
198 adonis2(m_trab ~ Periodo, data = t_grps, method = "bray")
199 adonis2(m_trab ~ Periodo*Rio, data = t_grps,
200         method = "bray") #+ no lugar de * tira os residuos
201 adonis2(m_trab ~ Periodo*Rio, data = t_grps,
202         method = "bray", by = "terms", strata = NULL)
203 sink()
204
205 all_p <- all_test$`Pr(>F)`[1]
206 all_p
207
208 count(t_grps, Periodo)
209 colnames(t_grps)
210
211 # Pairwise comparisons
212 #browseURL("https://doi.org/10.1007/s10641-018-0751-1")
213 #Lanés, L.E.K., Reichard, M., de Moura, R.G. et al. 2018.
214 #Environmental predictors for annual fish assemblages in subtropical
215 #grasslands of South America: the role of landscape and habitat
216 #characteristics. Environ Biol Fish 101, 963-977.
217
218 library(dplyr)
219 #library(combninat) # or base R combn()
220
221 set.seed(123)

```

```

222
223 # Get unique levels of UA
224 groups <- unique(t_grps$Ref_site)
225 groups
226 # All pairwise combinations
227 pairs <- combn(groups, 2, simplify = FALSE)
228 pairs
229 # Store results
230 pairwise_results <- lapply(pairs, function(g) {
231   # Subset data
232   sub_data <- t_grps %>% filter(Ref_site %in% g)
233   sub_dist <- m_trab[rownames(sub_data), ]
234
235   # Run adonis2
236   test <- adonis2(sub_dist ~ Ref_site, data = sub_data,
237     method = "bray", permutations = perm)
238
239   tibble(
240     group1 = g[1],
241     group2 = g[2],
242     F_value = test$F[1],
243     R2 = test$R2[1],
244     p_value = test$`Pr(>F)`[1]
245   )
246 })
247 # Bind all results
248 pairwise_results <- bind_rows(pairwise_results)
249 pairwise_results
250 pairwise_p <- numeric()
251 pairwise_p <- pairwise_results$p_value
252 pairwise_p
253 adj_p_values <- p.adjust(pairwise_p, method="bonferroni") #c("holm", "hochberg", "hommel", "
254 adj_p_values
255 pairwise_results$adj_p_values <- adj_p_values
256 pairwise_results
257 # Get unique group names
258 groups2 <- sort(unique(c(pairwise_results$group1, pairwise_results$group2)))
259 # Initialize empty matrix
260 pval_matrix <- matrix(NA, nrow = length(groups2), ncol = length(groups2),
261   dimnames = list(groups2, groups2))
262 # Fill in p-values
263 for (i in 1:nrow(pairwise_results)) {
264   g1 <- pairwise_results$group1[i]

```

```

265   g2 <- pairwise_results$group2[i]
266   p <- pairwise_results$p_value[i]
267
268   pval_matrix[g1, g2] <- p
269   pval_matrix[g2, g1] <- p #make symmetric
270 }
271 pval_matrix
272
273 library(knitr)
274 library(kableExtra)
275
276 # Make a copy of the matrix
277 pval_colored <- pval_matrix
278 # Replace numeric values with formatted HTML strings
279 pval_colored[] <- ifelse(is.na(pval_matrix), "",
280                          ifelse(pval_matrix < 0.05,
281                                paste0("<span style='color:red;'>", sprintf("%.3f", pval_matr
282                                sprintf("%.3f", pval_matrix)))
283 # Convert to data frame with rownames
284 pval_df <- as.data.frame(pval_colored)
285 pval_df <- tibble::rownames_to_column(pval_df, "Group")
286 # Render table
287 pval_df %>%
288   kable("html", escape = FALSE) %>%
289   kable_styling(full_width = FALSE)
290 save_kable(matrixstyle, "res/kb-perma_matrixstyle.html")
291
292 ## AVALIANDO INTERAÇÕES-----
293
294 adonis2(m_trab ~ Período, data = t_grps, method = "bray")
295 adonis2(m_trab ~ Período*Rio, data = t_grps,
296         method = "bray") #+ no lugar de * tira os residuos
297 adonis2(m_trab ~ Período*Rio, data = t_grps,
298         method = "bray", by = "terms", strata = NULL)

```

Among the averaged nutrient concentrations (Table 2), ammonia ranged from 49.2 to 243.7 $\mu\text{g L}^{-1}$. The Caiana stream tended to have greater values, but overall no clear trend can be observed when comparing both streams. Nitrite and nitrate, on the other hand, show clearly a greater concentrations at the Caiana stream, nitrate varying from 166.4 to 2003.0 $\mu\text{g L}^{-1}$, compared with the maximum of 18.8 $\mu\text{g L}^{-1}$ at the Barro Branco stream. Nitrite was too low to be detectable at the Barro Branco stream. The same pattern was observed for orthophosphate concentrations, being barely detectable at Barro Branco stream, and varying from 4.0 to 32.0 $\mu\text{g L}^{-1}$ at the Caiana stream. Total phosphorus, however, tended to have higher concentrations

at Barro Branco stream, reaching $50.1 \mu\text{g L}^{-1}$ at Site 3, but with highly variable values (from non detectable to $38.7 \mu\text{g L}^{-1}$, as the second highest value).

```
1  #TABELA DE NUTRIENTES-----
2
3  ##ORGANIZANDO DADOS-----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\014")
8
9  t_nutri <- read.csv("res/t_nutri.csv",
10                    sep = ";", dec = ".",
11                    row.names = 1,
12                    header = TRUE,
13                    na.strings = NA)
14
15  t_grps <- read.csv("res/t_grps.csv",
16                    sep = ";", dec = ".",
17                    row.names = 1,
18                    header = TRUE,
19                    na.strings = NA)
20
21  nutri <- t_nutri
22
23  library(dplyr)
24  library(tidyr)
25  t_grps <- t_grps[seq_len(nrow(nutri)), , drop = FALSE]
26  m_trab <- nutri %>%
27    rename_with(~ gsub("_", ".", .))
28  m <- m_trab %>%
29    group_by(PontoN = t_grps$PontoN) %>%
30    summarise(across(where(is.numeric),
31                      list(mean = mean, min = min, max = max)),
32              .groups = 'drop') %>%
33    pivot_longer(
34      cols = -c(PontoN),
35      names_to = c("Variable", ".value"),
36      names_sep = "_"
37    )
38
39  m <- as.data.frame(m)
40  colnames(m_trab)
```

```

41 m_wide <- m %>%
42   mutate(stat_string = ifelse(Variable == c("Ammonia.NH3.ug.L"),
43                               paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1),
44                               paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1),
45   unite("Location", PontoN, sep = "_") %>%
46   select(Variable, Location, stat_string) %>%
47   pivot_wider(names_from = Location, values_from = stat_string)
48
49 m_wide
50 m_wide <- as.data.frame(m_wide)
51 cols <- colnames(m_wide)
52 col_order <- c(
53   "Variable",
54   cols[cols != "Variable"][
55     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))]
56   ]
57 )
58 m_wide <- m_wide[, c(col_order)] #sequência das colunas
59 m_wide
60
61 # Salvado m_wide
62 write.table(m_wide, "res/m_wide_nutri.txt",
63             sep = ";", dec = ".", #"\t",
64             row.names = TRUE,
65             quote = TRUE,
66             append = FALSE)
67 m_wide_nutri <- read.table("res/m_wide_nutri.txt",
68                            sep = ";", dec = ".",
69                            row.names = 1,
70                            header = TRUE,
71                            na.strings = NA)
72
73 # Exportando dados para Excel
74 library(openxlsx)
75 #write.xlsx(m_wide, file = "res/tabela de nutrientes.xlsx", rowNames = FALSE)
76 wb <- loadWorkbook("res/tabela de nutrientes.xlsx")
77 writeData(wb, sheet = "Sheet 1", x = m_wide_nutri)
78 saveWorkbook(wb, "res/tabela de nutrientes.xlsx", overwrite = TRUE)
79
80 # Tabela final ajustada no Excel
81 library(readxl)
82 nutri_gtable <- read_excel(
83   path = "res/tabela de nutrientes.xlsx",

```

```

84   sheet = "tabela_final",
85   range = "A1:P7") #ajustar para o tamanho da tabela final
86 nutri_gtttable
87 library(gt)
88 nutri_gtttable <- gt(nutri_gtttable)
89 nutri_gtttable <- sub_missing(nutri_gtttable,
90                               columns = everything(), missing_text = "")
91 nutri_gtttable
92 gtsave(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.html")
93 saveRDS(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.rds")

```

3.2 Community structure

A total of 56 taxa distributed among Rotifera (14 families), Cladocera (5 families) and Copepoda (2 families) were identified during the study period. Rotifera was the most species-rich group, represented by 39 species (plus Bdelloidea), followed by Cladocera (12 species) and Copepoda (4 species, in addition to the juvenile stages nauplii and copepodite). Across sampling periods, Lecanidae exhibited the highest richness among all recorded families of Rotifera with 15 species, followed by Lepadellidae, Synchaetidae (both with 4 species), Brachionidae and Mytilinidae (both with 3 species). Bdelloidea was the most abundant taxa (6194 individuals) followed by *Lecane curvicornis* with 626 individuals. Among the cladocerans, Chydoridae was the richest family with 5 species, followed by Macrothricidae (3 species). *Alonella dadayi* was the most abundant among the cladocerans with 1051 individuals. Juvenile stages (nauplii and copepodite) were also abundant with 2433 and 4061 individuals, respectively. Copepods were represented by the orders Cyclopoida and Calanoida, *Paracyclops* sp. *Mesocyclops* sp. being the most abundant (141 and 231 individuals, respectively).

```

1  #ZOOEBIO_part - ESTR. DA COMUNIDADE----
2  #####....----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",

```

```

16         sep = ";", dec = ".",
17         row.names = 1,
18         header = TRUE,
19         na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21         sep = ";", dec = ".",
22         row.names = 1,
23         header = TRUE,
24         na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26         sep = ";", dec = ".",
27         row.names = 1,
28         header = TRUE,
29         na.strings = NA)
30
31 library(tibble)
32 library(tidyverse)
33 library(forcats)
34 library(openxlsx)
35 library(Rcpp)
36
37 m_dens
38 str(m_dens)
39 m_trab <- m_dens
40
41 m_trab <- (m_dens) #SITES
42 m_trab <- t(m_dens) #SPP
43
44 ##TAMANHO DA MATRIZ----
45
46 range(m_trab) #menor e maior valores
47 length(m_trab) #no. de colunas
48 ncol(m_trab) #no. de N colunas
49 nrow(m_trab) #no. de M linhas
50 sum(lengths(m_trab)) #soma os nos. de colunas
51 length(as.matrix(m_trab)) #tamanho da matriz m x n
52 sum(m_trab == 0) #número de observações igual a zero
53 sum(m_trab > 0) #número de observações maiores que zero
54 #calculando a proporção de zeros na matriz
55 zeros <- (sum(m_trab == 0)/length(as.matrix(m_trab)))*100
56 zeros
57
58 ##ESTRUTURA DA COMUNIDADE----

```

```

59
60 Sum <- rowSums(m_trab)
61 #ou
62 Sum <- apply(m_trab,1,sum)
63 Sum
64 # Abundância relativa (%)
65 RA <- (Sum / sum(Sum)) * 100 # percentage
66 # Media
67 Mean <- rowMeans(m_trab)
68 Mean
69 # Ou
70 Mean <- apply(m_trab,1,mean)
71 Mean
72 # Desvio padrão
73 DP <- apply(m_trab,1,sd)
74 DP
75 # Máximo
76 Max <- apply(m_trab,1,max)
77 Max
78 # Mínimo
79 Min <- apply(m_trab,1,min)
80 Min
81 # Mínimo não-zero
82 MinZ <- apply(m_trab, 1, function(row) {
83   non_zero_values <- row[row > 0] # Filter out zero values
84   if (length(non_zero_values) == 0) {
85     return(0) # If all values are zero, return 0
86   } else {
87     return(min(non_zero_values)) # Return the minimum of non-zero values
88   }
89 })
90 MinZ
91
92 m_pa <- m_trab
93 m_pa[m_pa != 0] <- 1
94 rowSums(m_pa)
95 library(vegan)
96 bin <- decostand(m_trab,"pa")
97 bin[1:10, 1:10]
98 S <- apply(bin,1,sum)
99 S
100 #OU
101 Riqueza <- specnumber(m_trab)

```

```

102 Riqueza
103 Riqueza_total <- specnumber(colSums(m_trab))
104 Riqueza_total
105
106 FO <- rowSums(m_trab > 0) / ncol(m_trab) * 100
107 FO
108
109 H <- diversity(m_trab, index = "shannon")
110 H
111 D <- diversity(m_trab, "simpson")
112 D
113 D[is.na(D)] <- 0 #substitui NA ou NaN por 0
114 D
115 E <- H/log(specnumber(m_trab))
116 E
117 E[is.na(E)] <- 0 #substitui NA ou NaN por 0
118 E
119
120 ###DESCRITORES----
121 Descritores1 <- cbind(Sum, RA, Mean, DP, Max, Min, MinZ, FO, S, E, H, D)
122 Descritores1 <- as.data.frame(Descritores1)
123 Descritores1
124 Descritores1 <- Descritores1[order(-Descritores1$Mean), ] #ordena do maior para o menor "-"
125 #Descritores1 <- Descritores1 %>% rownames_to_column(var="Espécies") #da nome a primeira col.
126 SomaTotalD <- apply(Descritores1,2,sum)
127 SomaTotalD
128 MediaTotalD <- apply(Descritores1,2,mean)
129 MediaTotalD
130 DPTotalD <- apply(Descritores1,2,sd)
131 DPTotalD
132 Descritores2 <- cbind(SomaTotalD, MediaTotalD, DPTotalD)
133 Descritores2 <- as.data.frame(Descritores2)
134 Descritores2 <- t(Descritores2)
135 Descritores2
136 DescritoresFinal <- rbind(Descritores1, Descritores2)
137 DescritoresFinal
138 DescritoresFinal <- round (DescritoresFinal, 2)
139 DescritoresFinal
140
141 # Salvando DescritoresFinal
142 write.table(data.frame("SPP"=rownames(DescritoresFinal),
143                       DescritoresFinal),
144             "res/DescritoresSPP.txt",

```

```

145         row.names=FALSE,
146         sep="\t")
147
148     ###TABELA FINAL DESCRITORES----
149     library(gt)
150     df <- DescritoresFinal
151     ncol(df); nrow(df) #no. de N colunas x M linhas
152     df <- cbind(SPP = rownames(df), df)
153     gt <- gt(df, rowname_col = "SPP",
154         caption = "Descritores da diversidade por espécie (colunas). Sum, soma; RA, abundância re
155     gt
156     gtsave(gt, "res/DescritoresSPP.html")
157     gtsave(gt, "res/DescritoresSPP.png")
158
159     ###NORMALIDADE----
160     library(moments)
161     Assimetria <- apply(m_trab,1,skewness)
162     Assimetria
163     Curtose <- apply(m_trab,1,kurtosis)
164     Curtose
165     Normalidade1 <- cbind(Assimetria, Curtose)
166     Normalidade1 <- as.data.frame(Normalidade1)
167     Normalidade1
168     SomaTotalN <- apply(Normalidade1,2,sum)
169     SomaTotalN
170     MediaTotalN <- apply(Normalidade1,2,mean)
171     MediaTotalN
172     DPTotalN <- apply(Normalidade1,2,sd)
173     DPTotalN
174     Normalidade2<-cbind(SomaTotalN, MediaTotalN, DPTotalN)
175     Normalidade2<-as.data.frame(Normalidade2)
176     Normalidade2 <- t(Normalidade2) #"t" transpoe a matriz
177     Normalidade2
178     NormalidadeFinal <- rbind(Normalidade1, Normalidade2)
179     NormalidadeFinal
180     NormalidadeFinal <- round(NormalidadeFinal, 2)
181     NormalidadeFinal
182
183     # Salvando NormalidadeFinal
184     write.table(data.frame("SPP"=rownames(NormalidadeFinal),
185         NormalidadeFinal),
186         "res/NormalidadeSPP.txt",
187         row.names=FALSE,

```

```

188         sep="\t")
189
190 ###TABELA FINAL NORMALIDADE----
191 nf <- NormalidadeFinal
192 ncol(nf); nrow(nf) #no. de N colunas x M linhas
193 nf <- cbind(SPP = rownames(nf), nf)
194 gt <- gt(nf, rowname_col = "SPP", caption = "Descritores da normalidade por SPP (COLUNA)")
195 gt
196 gtsave(gt, "res/NormalidadeSPP.html")
197
198 #DIVERSIDADE EM DENSIDADE MATRIZ NÃO-TRANSPOSTA----
199
200 m_trab <- (m_dens) #HILL
201
202 # Números de Hill
203 Hill <- renyi(m_trab,
204             scales = c(0:5),
205             hill = TRUE)
206 Hill
207 library(ggplot2)
208 library(tidyr)
209 library(tidyverse)
210 grafico1 <- Hill %>%
211   rownames_to_column() %>%
212   pivot_longer(-rowname) %>%
213   mutate(name = factor(name, name[1:length(Hill)])) %>%
214   ggplot(aes(x = name, y = value, group = rowname,
215             col = rowname)) +
216   geom_point(size = 2) +
217   geom_line(size = 1) +
218   xlab("Parâmetro de ordem de diversidade (q)") +
219   ylab("Diversidade") +
220   labs(col = "Locais") +
221   theme_bw() +
222   theme(text = element_text(size = 10)) #ajustar a fonte caso nao caiba no output html
223 grafico1
224 ggsave(grafico1, dpi = 300, filename = "res/fig-hill.png")
225
226 # Gráfico de abundância
227 abund <- colSums(m_trab)
228 #abund <- m_trab[1, ] #escolhe a primeira linha para a distribuição de abundância
229 df <- data.frame(sp = colnames(m_trab),
230                 abun = abund)

```

```

231 grafico2 <- ggplot(df, aes(fct_reorder(sp, -abun),
232                               abun, group = 1)) +
233   geom_col() +
234   geom_line(col = "red", linetype = "dashed") +
235   geom_point(col = "red") +
236   xlab("Espécies") +
237   ylab("Abundância") +
238   theme_bw() +
239   theme(axis.text.x = element_text(
240     angle = 45,
241     hjust = 1,
242     face = "italic",
243     size = 10),
244     axis.text.y = element_text(size = 10),
245     axis.title.x = element_text(size = 12),
246     axis.title.y = element_text(size = 12))
247 grafico2
248 ggsave(grafico2, dpi = 300, filename = "res/fig-abun.png")
249
250 # Curvas de rarefação
251 m_trab_r <- mutate(as.data.frame(m_trab), across(everything(), ceiling)) #arredonda a matriz
252 colSums(t(m_trab_r))
253 colnames(t(m_trab_r))
254 #rarefa <- t(m_trab_r[c("S-R-CT2", "S-R-CP2", "S-A-TA2", "B-A-MU2"),]) #curva de rarefação para
255 class(m_trab)
256
257 # Curva de rarefação para todos os sítios
258 rarefa <- t(m_trab_r)
259 ## Curva de rarefação para as 8 UA's com maior soma
260 #m_trab_r <- as.data.frame(t(m_trab_r)) #transpõe a matriz
261 #col_sums <- colSums(m_trab_r)
262 #largest_columns <- names(sort(col_sums, decreasing = TRUE)[1:8])
263 #rarefa <- m_trab_r[largest_columns] #curva de rarefação para as 8 UA's com maior soma
264 library(iNEXT)
265 out <- iNEXT(rarefa, q = 0,
266             datatype = "abundance",
267             size = NULL,
268             endpoint = 2000, #define o comprimento de eixo x
269             knots = 40,
270             se = TRUE,
271             conf = 0.95,
272             nboot = 50)
273 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") +

```

```

274   theme_bw() +
275   labs(fill = "Áreas") +
276   xlab("Número de indivíduos") +
277   ylab("Riqueza de espécies") +
278   theme(legend.title=element_blank()) #ver como fica com facet.var="Assemblage"
279   #grafico3
280   #ggsave(grafico3, dpi = 300, filename = "fig-rare1.png")
281
282   # Curvas de acumulação de espécies
283   acumula <- specaccum(m_trab,
284                       method = "random")
285   acumula
286   plot(acumula)
287   plot_data <- data.frame("UAs" = c(0, acumula$sites),
288                          "Riqueza" = c(0, acumula$richness),
289                          "lower" = c(0, acumula$richness - acumula$sd),
290                          "upper" = c(0, acumula$richness + acumula$sd))
291   gLocais <- ggplot(plot_data, aes(x = UAs, y = Riqueza)) +
292     geom_point(color = "blue", size = 4) +
293     geom_line(color = "blue", lwd = 2) +
294     geom_ribbon(aes(ymin = lower, ymax = upper),
295               linetype=2, alpha=0.3, fill = "yellow") +
296     ylab("Riqueza acumulada") +
297     theme_classic() +
298     theme(text = element_text(size = 16))
299   gLocais
300   ggsave(gLocais, dpi = 300, filename = "res/fig-acum.png")
301
302   plot_data <- data.frame("Individuos" = c(0, acumula$individuals),
303                          "Riqueza" = c(0, acumula$richness),
304                          "lower" = c(0, acumula$richness - acumula$sd),
305                          "upper" = c(0, acumula$richness + acumula$sd))
306   gInd <- ggplot(plot_data, aes(x = Individuos, y = Riqueza)) +
307     geom_point(color = "blue", size = 4) +
308     geom_line(color = "blue", lwd = 2) +
309     geom_ribbon(aes(ymin = lower, ymax = upper),
310               linetype=2, alpha=0.3, fill = "yellow") +
311     ylab("Riqueza acumulada") +
312     theme_classic() +
313     theme(text = element_text(size = 16))
314   gInd
315
316   #RIQUEZA RAREFACIONADA----

```

```

317
318 # Rarefação baseada na matriz de contagem
319 library(vegan)
320 data <- m_cont #raremax precisa de numeros inteiros
321 S <- specnumber(data) #observed number of species
322 S
323 #raremax <- min(rowSums(data))
324 raremax <- round(mean(rowSums(data)))
325 #raremax <- 1000
326 raremax
327 Srare <- rarefy(data, raremax)
328 Srare
329 plot(S, Srare, xlab = "Observed No. of Species", ylab = "Rarefied No. of Species")
330 abline(0, 1)
331
332 #png("res/fig-rarecurve.png")
333 rarecurve(data, step = 20,
334           sample = raremax, #linhas de rarefação
335           col = "blue", cex = 0.6,
336           xlab = "Sample size", ylab = "Taxa", xlim = c()) #c(0,5000)
337 #dev.off()
338
339 ##TABELA DA COMUNIDADE----
340
341 library(dplyr)
342 library(tidyr)
343 m_trab <- m_dens %>%
344   rename_with(~ gsub("_", ".", .))
345 m <- m_trab %>%
346   group_by(Rio = t_grps$Rio,
347            Periodo = t_grps$Periodo,
348            Ponto = t_grps$PontoN) %>%
349   summarise(across(where(is.numeric),
350                    list(mean = mean, sd = sd)),
351            .groups = 'drop') %>%
352   pivot_longer(
353     cols = -c(Rio, Periodo, Ponto),
354     names_to = c("Variable", ".value"),
355     names_sep = "_"
356   )
357 m
358 m <- as.data.frame(m)
359 m_wide <- m %>%

```

```

360 mutate(stat_string = ifelse(
361   Variable == "nome_da_variavel", #escolhe uma variável pra ter 3 casas decimais
362   paste0(round(mean, 3), " (", round(sd, 3), ")"),
363   paste0(round(mean, 1), " (", round(sd, 1), ")")
364 )) %>%
365 unite("Location", Rio, Período, Ponto, sep = "_") %>%
366 dplyr::select(Variable, Location, stat_string) %>% #dplyr dá conflito com MASS
367 pivot_wider(names_from = Location, values_from = stat_string)
368
369 m_wide
370 m_wide <- as.data.frame(m_wide)
371 m_wide
372
373 # Salvado m_wide
374 write.table(m_wide, "res/m_wide_spp.txt",
375             sep = ";", dec = ".", #"\\t",
376             row.names = TRUE,
377             quote = TRUE,
378             append = FALSE)
379 m_wide_spp <- read.table("res/m_wide_spp.txt",
380                          sep = ";", dec = ".",
381                          row.names = 1,
382                          header = TRUE,
383                          na.strings = NA)
384
385 # Exportando dados para Excel
386 library(openxlsx)
387 #write.xlsx(m_wide_spp, file = "res/tabela da comunidade.xlsx", rowNames = FALSE)
388 wb <- loadWorkbook("res/tabela da comunidade.xlsx")
389 writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
390 saveWorkbook(wb, "res/tabela da comunidade.xlsx", overwrite = TRUE)
391
392 # Tabela final ajustada no Excel
393 library(readxl)
394 spp_gtable <- read_excel(
395   path = "res/tabela da comunidade.xlsx",
396   sheet = "tabela_final",
397   col_names = T,
398   range = "A1:AE61") #ajustar para o tamanho da tabela final
399 spp_gtable
400 library(gt)
401 spp_gtable <- gt(spp_gtable)
402 spp_gtable <- sub_missing(spp_gtable,

```

```

403         columns = everything(), missing_text = "")
404 #spp_gtttable <- fmt_number(spp_gtttable,
405 #                           columns = "F.O.(%)", decimals = 1)
406 spp_gtttable
407 gtsave(spp_gtttable, "res/gt-spp_gtttable_xlsx.html")
408 saveRDS(spp_gtttable, "res/gt-spp_gtttable_xlsx.rds")

1  # Escolher sumário de uma variável
2  m
3  var <- "h.roots"
4  m[m$Variable == var, "mean"] #cada valor de var
5  summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7  # Escolher sumário de um grupo de variáveis do df m
8  vars <- unique(grep("^h\\.", m$Variable, value = TRUE))
9  summaries <- list() #cria uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL
27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

Rarefaction analysis showed that the Caiana stream had a taxa richness slightly higher than that of the Barro Branco stream (Figure 4). Nevertheless, the Chao index estimated an extrapolated richness of 58 (± 8.9) species for the Barro Branco stream, compared with 54.2 (± 4.3) species for the Caiana stream.

```

1  #ZOOREBIO_part - CURVA DE ACUMULAÇÃO----
2  #####....----

```

```

3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_cont <- read.csv("res/m_cont.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_trab <- m_cont
16
17 library(tidyverse)
18
19 m_trab_r <- mutate(m_trab, across(everything(), ceiling)) #arredonda a matriz original para
20
21 colSums(t(m_trab_r))
22 colnames(t(m_trab_r))
23
24 ca_abund <- colSums(m_trab_r[grepl("-C-", rownames(m_trab_r)), ])
25 bb_abund <- colSums(m_trab_r[grepl("-B-", rownames(m_trab_r)), ])
26
27 m_rios <- rbind("Caiana" = ca_abund, "Barro Branco" = bb_abund)
28 m_rios
29
30 rarefa <- t(m_rios)
31
32 library(iNEXT)
33 out <- iNEXT(rarefa, q = 0,
34             datatype = "abundance",
35             size = NULL,
36             endpoint = 15000, #define o comprimento de eixo x
37             knots = 40,
38             se = TRUE,
39             conf = 0.95,
40             nboot = 50)
41
42 library(ggplot2)
43 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") + #ver como fica com facet.var="Assembl
44   theme_bw() +
45   labs(fill = "Áreas") +

```

```

46   xlab("Número de individuos") +
47   ylab("Riqueza de especies") +
48   scale_linetype_discrete(
49     labels = c("Interpolation", "Extrapolation")) +
50   theme(legend.title = element_blank(),
51         legend.position = "bottom",
52         legend.text = element_text(size = 14),
53         axis.title = element_text(size = 14),
54         axis.text = element_text(size = 12)
55   )
56   grafico3
57   ggsave(grafico3, width = 8, height = 6, dpi = 300, filename = "res/fig-rare1.png")
58
59   #ESTIMADORES DA DIVERSIDADE
60
61   #https://rpubs.com/Juan-Mt/diversity-estimations
62   #https://analises-ecologicas.com/cap11
63
64   m_cont <- as.matrix(m_cont)
65   storage.mode(m_cont) <- "numeric"
66
67   BB <- m_cont[grepl("^[SC]-B-", rownames(m_cont)),, drop = FALSE]
68   CA <- m_cont[grepl("^[SC]-C-", rownames(m_cont)),, drop = FALSE]
69
70   nrow(BB)
71   nrow(CA)
72
73   # Calculando a riqueza observada
74
75   BB_total <- colSums(BB)
76   CA_total <- colSums(CA)
77
78   S_BB <- sum(BB_total > 0)
79   S_CA <- sum(CA_total > 0)
80
81   S_BB
82   S_CA
83
84   # Chao1
85   chao_BB <- estimateR(BB_total)
86   chao_CA <- estimateR(CA_total)
87
88   chao_BB

```

```

89  chao_CA
90
91  # Separando Chao1
92  resultado_chao <- data.frame(
93    Stream = c("Barro Branco", "Caiana"),
94    Observed = c(
95      chao_BB["S.obs"],
96      chao_CA["S.obs"]
97    ),
98    Chao1 = c(
99      chao_BB["S.chao1"],
100     chao_CA["S.chao1"]
101   ),
102   SE = c(
103     chao_BB["se.chao1"],
104     chao_CA["se.chao1"]
105   )
106 )
107 resultado_chao
108
109 # 95% confidence intervals
110 resultado_chao$Lower95 <- resultado_chao$Chao1 -
111   1.96 * resultado_chao$SE
112 resultado_chao$Upper95 <- resultado_chao$Chao1 +
113   1.96 * resultado_chao$SE
114
115 # Do not allow the lower limit to be below observed richness
116 resultado_chao$Lower95 <- pmax(
117   resultado_chao$Lower95,
118   resultado_chao$Observed
119 )
120
121 sink(file = "res/chao.txt")
122 # Tabela pronta
123 resultado_chao
124 sink()

```

The NMDS ordination diagrams showed that the composition of zooplankton resting stages differed among most of the ponds from the forest (Fig. 4A) and grassland biomes (Fig. 4B). In addition, the PERMANOVA indicated a significant difference in the composition of zooplanktonic taxa between biomes (pseudo-F_{1, 53} = 1.85, R² = 0.035; P = 0.03). The outputs of the db-RDAs showed that the water chemistry subset significantly influenced the dissimilarity in the resting stage composition (F = 1.25; R² = 0.134; P = 0.02), while the subset of habitat

structure did not ($F = 0.88$; $R^2 = 0.01$; $P = 0.88$, Fig. 5).

The UPGMA cluster analyses revealed low-to-moderate spatial clustering strength for the zooplankton communities across both periods, as indicated by the Agglomerative Coefficients ($AC_{dry} = 0.34$ and $AC_{wet} = 0.39$). While community composition exhibited continuous variation across sites, spatial structuring was slightly more pronounced during the wet season than during the dry season.

```
1  #ZOOEBIO_part - UPGMA----
2  #####.....-----
3
4  ##ORGANIZANDO DADOS-----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                   sep = ";", dec = ".",
17                   row.names = 1,
18                   header = TRUE,
19                   na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21                       sep = ";", dec = ".",
22                       row.names = 1,
23                       header = TRUE,
24                       na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26                   sep = ";", dec = ".",
27                   row.names = 1,
28                   header = TRUE,
29                   na.strings = NA)
30
31 m_dens
32 str(m_dens)
33
34 #SEPARANDO OS PERÍODOS
35
```

```

36 library(vegan)
37 library(cluster)
38
39 # Fazendo a média por ponto
40 ponto_id <- sub("-H[0-9]+$", "", rownames(m_dens))
41 m_dens_media <- aggregate(m_dens, by = list(Ponto = ponto_id), FUN = mean)
42 rownames(m_dens_media) <- m_dens_media$Ponto
43 m_dens_media$Ponto <- NULL
44 # Separar por período
45 seco_med <- m_dens_media[startsWith(rownames(m_dens_media), "S-"), ]
46 chuv_med <- m_dens_media[startsWith(rownames(m_dens_media), "C-"), ]
47 # Filtrar COLUNAS zeradas (espécies ausentes no período)
48 seco_med <- seco_med[, colSums(seco_med) > 0]
49 chuv_med <- chuv_med[, colSums(chuv_med) > 0]
50 # Filtrar LINHAS zeradas (pontos sem organismos)
51 seco_med <- seco_med[rowSums(seco_med) > 0, ]
52 chuv_med <- chuv_med[rowSums(chuv_med) > 0, ]
53 # Transformação
54 seco_trns <- asin(sqrt(decostand(seco_med,
55                               method="total", MARGIN = 2)))
56 chuv_trns <- asin(sqrt(decostand(chuv_med,
57                               method="total", MARGIN = 2)))
58 # Distâncias de Bray-Curtis e UPGMA
59 dist_seco <- vegdist(seco_trns, method = "bray")
60 dist_chuv <- vegdist(chuv_trns, method = "bray")
61 upgma_seco <- hclust(dist_seco, method = "average")
62 upgma_chuv <- hclust(dist_chuv, method = "average")
63
64 # Coeficientes de aglomeração
65 agg_coef_seco <- coef.hclust(upgma_seco)
66 agg_coef_chuv <- coef.hclust(upgma_chuv)
67 # Gerar a matriz de distância cofenética a partir da árvore UPGMA
68 cofe_seco <- cophenetic(upgma_seco)
69 cofe_chuv <- cophenetic(upgma_chuv)
70 # Calcular a correlação com a matriz de distância original (Bray-Curtis)
71 cor_seco <- cor(dist_seco, cofe_seco)
72 cor_chuv <- cor(dist_chuv, cofe_chuv)
73 # Visualizar os resultados
74 cat("Correlação Cofenética (Seco):", round(cor_seco, 3), "\n")
75 cat("Correlação Cofenética (Chuvoso):", round(cor_chuv, 3), "\n")
76
77 # Plot dos dendrogramas agrupados por ponto
78 upgma_seco$labels <- sub("[A-Z]-", "", upgma_seco$labels)

```

```

79 upgma_chuv$labels <- sub("[A-Z]-", "", upgma_chuv$labels)
80 dend_seco <- as.dendrogram(hclust(dist_seco, method = "average"))
81 dend_chuv <- as.dendrogram(hclust(dist_chuv, method = "average"))
82
83 #png("res/fig-upgma.png", width=11, height=9, units="in", res=400)
84 par(mfrow = c(1, 2), mar = c(5, 4, 2, 6)) #B, L, T, R
85 plot(as.dendrogram(upgma_seco),
86       horiz = TRUE,
87       xlab = "Bray-Curtis dissimilariry",
88       xlim = c(max(attr(dend_seco, "height")), 0),
89       cex.lab = 1)
90 title(main = "Dry", line = 0, cex.main = 1.2)
91 mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_seco, 2)),
92       side = 2,           # Lado esquerdo (eixo Y)
93       line = 1,           # Distância da margem
94       at = 7,             # Posição na base do eixo
95       las = 0,            # Orientação do texto
96       cex = 0.8)          # Tamanho da fonte
97 mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
98       side = 2,           # Lado esquerdo (eixo Y)
99       line = 2,           # Distância da margem
100      at = 7,             # Posição na base do eixo
101      las = 0,            # Orientação do texto
102      cex = 0.8)          # Tamanho da fonte
103
104 plot(as.dendrogram(upgma_chuv),
105       horiz = TRUE,
106       xlab = "Bray-Curtis dissimilariry",
107       xlim = c(max(attr(dend_chuv, "height")), 0),
108       cex.lab = 1)
109 title(main = "Wet", line = 0, cex.main = 1.2)
110 mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
111       side = 2,           # Lado esquerdo (eixo Y)
112       line = 1,           # Distância da margem
113       at = 7,             # Posição na base do eixo
114       las = 0,            # Orientação do texto
115       cex = 0.8)          # Tamanho da fonte
116 mtext(paste0("Cophenetic Coefficient = ", round(cor_chuv, 2)),
117       side = 2,           # Lado esquerdo (eixo Y)
118       line = 2,           # Distância da margem
119       at = 7,             # Posição na base do eixo
120       las = 0,            # Orientação do texto
121       cex = 0.8)          # Tamanho da fonte

```

```

122 par(mfrow = c(1, 1))
123 #dev.off()
124
125 #OUTRA FORMA
126 #plot(as.dendrogram(upgma_chuv),
127 #      horiz = TRUE,
128 #      main = "Wet",
129 #      xlab = "Bray-Curtis dissimilarity",
130 #      xlim = c(max(attr(dend_chuv, "height")), 0))
131 # Adiciona o texto no canto inferior esquerdo
132 #text(x = max(attr(dend_chuv, "height")) * 0.9,
133 #      y = 3,
134 #      labels = paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
135 #      adj = c(0, 0),
136 #      cex = 0.85,
137 #      srt = 90) # srt = 90 gira o texto para ficar vertical
138 #OU
139 #mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
140 #      side = 2,      # 1 = bottom margin
141 #      line = 2.5,    # Line distance below axis (increase if it overlaps numbers)
142 #      adj = 0.36,    # Posição horizontal
143 #      cex = 0.85) # Font size

```

3.3 Diversity Partitioning

```

1  #ZOOEBIO_part - ADDEPART----
2  #####....-----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_esc <- read.csv("res/m_esc.csv",
11                  sep = ";", dec = ".",
12                  row.names = 1,
13                  header = TRUE,
14                  na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                  sep = ";", dec = ".",
17                  row.names = 1,

```

```

18         header = TRUE,
19         na.strings = NA)
20 m_cont <- read.csv("res/m_cont.csv",
21                 sep = ";", dec = ".",
22                 row.names = 1,
23                 header = TRUE,
24                 na.strings = NA)
25
26 #BUSCANDO AS ESCALAS
27
28 # Partição total
29 colnames(m_esc)
30 pt.total <- m_esc[, c("pt.rio", "pt.periodo", "pt.rebio")]
31 head(pt.total)
32 dim(pt.total)
33
34 # Partição seco
35 pt.seco <- m_esc[, c("s.habitat", "s.ponto", "s.rio", "s.rebio")]
36 pt.seco <- pt.seco[grepl("^S-", rownames(pt.seco)), ]
37 head(pt.seco)
38 dim(pt.seco)
39 m_cont_s <- m_cont[grepl("^S-", rownames(m_cont)), ]
40
41 # Partição chuvoso
42 pt.chuva <- m_esc[, c("c.habitat", "c.ponto", "c.rio", "c.rebio")]
43 pt.chuva <- pt.chuva[grepl("^C-", rownames(pt.chuva)), ]
44 head(pt.chuva)
45 dim(pt.chuva)
46 m_cont_c <- m_cont[grepl("^C-", rownames(m_cont)), ]
47
48 # Partição bb.seco (A)
49 bb.seco <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
50 bb.seco <- bb.seco[grepl("S-B-", rownames(bb.seco)), ]
51 head(bb.seco)
52 dim(bb.seco)
53 m_cont_bb.seco <- m_cont[grepl("S-B-", rownames(m_cont)), ]
54
55 # Partição bb.chuva (B)
56 bb.chuva <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
57 bb.chuva <- bb.chuva[grepl("C-B-", rownames(bb.chuva)), ]
58 head(bb.chuva)
59 dim(bb.chuva)
60 m_cont_bb.chuva <- m_cont[grepl("C-B-", rownames(bb.chuva)), ]

```

```

61
62 # Partição ca.seco (C)
63 ca.seco <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
64 ca.seco <- ca.seco[grep("S-C-", rownames(ca.seco)), ]
65 head(ca.seco)
66 dim(ca.seco)
67 m_cont_ca.seco <- m_cont[grep("S-C-", rownames(ca.seco)), ]
68
69 # Partição ca.chuva (D)
70 ca.chuva <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
71 ca.chuva <- ca.chuva[grep("C-C-", rownames(ca.chuva)), ]
72 head(ca.chuva)
73 dim(ca.chuva)
74 m_cont_ca.chuva <- m_cont[grep("C-C-", rownames(ca.chuva)), ]
75
76 #PARTIÇÃO
77
78 #pt.seco (m_cont_s), pt.chuva (m_cont_c)
79 #bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
80 #ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)
81
82 part1 <- ca.chuva
83 cont <- m_cont_ca.chuva
84
85 library(vegan)
86
87 sink(file = "res/part-ca.chuva.txt")
88 #VALORES ABSOLUTOS
89 part.a <- adipart(y = cont, x = part1, index = "richness", relative = FALSE,
90                 weights = "unif", nsimul = 999)
91 part.a
92
93 #EM PORCENTAGEM
94 part.p <- adipart(y = cont, x = part1, index = "richness", relative = TRUE,
95                 weights = "unif", nsimul = 999)
96 part.p
97
98 #str(part.p)
99
100 # Extrair valores do objeto part.p
101
102 part2 <- part.p
103

```

```

104 statistic <- part2$statistic
105 SES <- part2$oecosimu$z
106 mean <- part2$oecosimu$means
107 lower_2.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.025)
108 median_50 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.50)
109 upper_97.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.975)
110 Pr_sim <- part2$oecosimu$pval
111
112 # Definir função para adicionar significância
113 get_significance <- function(p) {
114   if (p <= 0.001) {
115     return("****")
116   } else if (p <= 0.01) {
117     return("***")
118   } else if (p <= 0.05) {
119     return("**")
120   } else if (p <= 0.1) {
121     return(".")
122   } else {
123     return(" ")
124   }
125 }
126 significance <- sapply(Pr_sim, get_significance)
127
128 # Criar dataframe com todas as informações
129
130 diversity_data <- data.frame(
131   Component = names(statistic),
132   Observed = round(statistic, 2),
133   SES = round(SES, 2),
134   Mean = round(mean, 2),
135   Lower_2.5 = round(lower_2.5, 2),
136   Median_50 = round(median_50, 2),
137   Upper_97.5 = round(upper_97.5, 2),
138   Pr_sim = Pr_sim,
139   Sign = significance
140 )
141
142 # Exibir a tabela
143 library(gt)
144 print(diversity_data)
145 #str(part2)
146 gt(diversity_data)

```

```

147
148 # Criar dataframe com valores observados e esperados
149 observed_values <- part2$statistic
150 observed_values
151 expected_values <- part2$oecosimu$means
152 expected_values
153
154 # Combine observed and expected values into a dataframe
155 df <- data.frame(
156   Component = rep(names(observed_values), 2), # Use names from observed_values
157   Type = rep(c("Observed", "Expected"), each = length(observed_values)),
158   Value = c(observed_values, expected_values) # Observado primeiro, Esperado depois
159 )
160
161 df$Type <- factor(df$Type, levels = c("Observed", "Expected"))
162 df
163 df_filtered <- df[!df$Component %in% c("alpha.2", "alpha.3", "gamma"), ]
164 df_filtered$Component <- factor(df_filtered$Component, levels = c("beta.3", "beta.2", "beta.1"))
165 df_filtered
166
167 diversity_data <- df_filtered
168 diversity_data
169
170 # Calcular porcentagens para cada componente
171 diversity_data <- diversity_data %>%
172   group_by(Type) %>%
173   mutate(Percentage = Value / sum(Value) * 100)
174
175 hier <- hiersimu(y = cont, x = part1, FUN = diversity)
176 hier
177 sink()
178
179 #GRÁFICOS
180
181 library(ggplot2)
182 library(dplyr)
183
184 # Partição total
185
186 ggplot_pt.total <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
187   geom_bar(position = "stack", stat = "identity") +
188   labs(title = "Total",
189        y = "% of the total diversity",

```

```

190     x = "") +
191     scale_fill_manual(values = c("alpha.1" = "black",
192                                "beta.1" = "darkgray",
193                                "beta.2" = "gray",
194                                "beta.3" = "lightgray")) +
195     theme_minimal() +
196     theme(
197       axis.text.x = element_text(angle = 0, hjust = 0.5),
198       panel.grid = element_blank(),
199     )
200 ggplot_pt.total
201 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_total.png")
202
203 # Partição seco
204
205 ggplot_pt.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
206   geom_bar(position = "stack", stat = "identity") +
207   labs(title = "Seco",
208        y = "% of the total diversity",
209        x = "") +
210   scale_fill_manual(values = c("alpha.1" = "black",
211                                "beta.1" = "darkgray",
212                                "beta.2" = "gray",
213                                "beta.3" = "lightgray")) +
214   theme_minimal() +
215   theme(
216     axis.text.x = element_text(angle = 0, hjust = 0.5),
217     panel.grid = element_blank(),
218   )
219 ggplot_pt.seco
220 ggsave(ggplot_pt.seco, dpi = 300, filename = "res/fig-part_seco.png")
221
222 # Partição chuvoso
223
224 ggplot_pt.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
225   geom_bar(position = "stack", stat = "identity") +
226   labs(title = "Chuvoso",
227        y = "% of the total diversity",
228        x = "") +
229   scale_fill_manual(values = c("alpha.1" = "black",
230                                "beta.1" = "darkgray",
231                                "beta.2" = "gray",
232                                "beta.3" = "lightgray")) +

```

```

233   theme_minimal() +
234   theme(
235     axis.text.x = element_text(angle = 0, hjust = 0.5),
236     panel.grid = element_blank(),
237   )
238 ggplot_pt.chuva
239 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_chuva.png")
240
241 # bb.seco (A)
242
243 ggplot_bb.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
244   geom_bar(position = "stack", stat = "identity") +
245   labs(title = "(A)",
246     y = "% of the total diversity",
247     x = "") +
248   scale_fill_manual(values = c("alpha.1" = "black",
249     "beta.1" = "darkgray",
250     "beta.2" = "gray",
251     "beta.3" = "lightgray")) +
252   theme_minimal() +
253   theme(
254     axis.text.x = element_text(angle = 0, hjust = 0.5),
255     panel.grid = element_blank(),
256   )
257 ggplot_bb.seco
258 ggsave(ggplot_bb.seco, dpi = 300, filename = "res/fig-bb.seco.png")
259
260 # bb.chuva (B)
261
262 ggplot_bb.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
263   geom_bar(position = "stack", stat = "identity") +
264   labs(title = "(B)",
265     y = "% of the total diversity",
266     x = "") +
267   scale_fill_manual(values = c("alpha.1" = "black",
268     "beta.1" = "darkgray",
269     "beta.2" = "gray",
270     "beta.3" = "lightgray")) +
271   theme_minimal() +
272   theme(
273     axis.text.x = element_text(angle = 0, hjust = 0.5),
274     panel.grid = element_blank(),
275   )

```

```

276 ggplot_bb.chuva
277 ggsave(ggplot_bb.chuva, dpi = 300, filename = "res/fig-bb.chuva.png")
278
279 # ca.seco (C)
280
281 ggplot_ca.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
282   geom_bar(position = "stack", stat = "identity") +
283   labs(title = "(C)",
284        y = "% of the total diversity",
285        x = "") +
286   scale_fill_manual(values = c("alpha.1" = "black",
287                                "beta.1" = "darkgray",
288                                "beta.2" = "gray",
289                                "beta.3" = "lightgray")) +
290   theme_minimal() +
291   theme(
292     axis.text.x = element_text(angle = 0, hjust = 0.5),
293     panel.grid = element_blank(),
294   )
295 ggplot_ca.seco
296 ggsave(ggplot_ca.seco, dpi = 300, filename = "res/fig-ca.seco.png")
297
298 # ca.chuva (D)
299
300 ggplot_ca.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
301   geom_bar(position = "stack", stat = "identity") +
302   labs(title = "(D)",
303        y = "% of the total diversity",
304        x = "") +
305   scale_fill_manual(values = c("alpha.1" = "black",
306                                "beta.1" = "darkgray",
307                                "beta.2" = "gray",
308                                "beta.3" = "lightgray")) +
309   theme_minimal() +
310   theme(
311     axis.text.x = element_text(angle = 0, hjust = 0.5),
312     panel.grid = element_blank(),
313   )
314 ggplot_ca.chuva
315 ggsave(ggplot_ca.chuva, dpi = 300, filename = "res/fig-ca.chuva.png")
316
317 # PAINEL DE GRÁFICOS
318

```

```

319 library(gridExtra)
320
321 # 1. Organizar os gráficos em 2 linhas e 2 colunas
322
323 painel_completo <- grid.arrange(
324   ggplot_bb.seco, ggplot_bb.chuva,
325   ggplot_ca.seco, ggplot_ca.chuva,
326   ncol = 2, nrow = 2
327 )
328
329 # 2. Salvar o painel em alta resolução (300 DPI)
330 ggsave(
331   filename = "res/fig-painel_partição.png",
332   plot = painel_completo,
333   width = 10,
334   height = 8,
335   dpi = 300
336 )

```

1 #TABELA DE PARTIÇÃO----

2 #Feita no Gemini, conferir

3

4 # -----

5 # 1. FUNÇÃO DE EXTRAÇÃO

6 # -----

```

7 extrair_linha_particao <- function(part_obj, sistema, periodo) {
8
9   mapa_componentes <- c(
10     "alpha.1" = " 1 (micro-habitat)",
11     "alpha.2" = " 2 (ponto)",
12     "alpha.3" = " 3 (riacho)",
13     "beta.1"  = " 1 (entre micro-habitats)",
14     "beta.2"  = " 2 (entre pontos)",
15     "beta.3"  = " 3 (entre riachos)",
16     "gamma"   = " (Riqueza total)"
17   )
18
19   stat <- part_obj$statistic
20   comp_names <- names(stat)
21
22   df <- data.frame(
23     Sistema    = sistema,
24     Período    = periodo,

```

```

25     Componente = ifelse(comp_names %in% names(mapa_componentes),
26                         mapa_componentes[comp_names], comp_names),
27     Observado = sprintf("%.2f%%", stat * 100),
28     Esperado = sprintf("%.2f%%", part_obj$oecosimu$means * 100),
29     SES      = sprintf("%.2f", part_obj$oecosimu$z),
30     p_val    = part_obj$oecosimu$pval,
31     stringsAsFactors = FALSE
32 )
33
34 df$p_valor <- ifelse(df$p_val <= 0.001, "< 0.001*",
35                     ifelse(df$p_val <= 0.05, sprintf("%.3f*", df$p_val),
36                     sprintf("%.3f", df$p_val)))
37
38 return(df %>% select(-p_val))
39 }
40
41 # -----
42 # 2. PROCESSAMENTO E RUNDOWN DOS OBJETOS ADIPART
43 # -----
44 # Rodando o adipart para cada cenário e gerando os objetos corretos
45
46 p_ca_seco <- adipart(y = m_cont_ca.seco, x = ca.seco, index = "richness", relative = TRUE)
47 p_ca_chuva <- adipart(y = m_cont_ca.chuva, x = ca.chuva, index = "richness", relative = TRUE)
48
49 p_bb_seco <- adipart(y = m_cont_bb.seco, x = bb.seco, index = "richness", relative = TRUE)
50 p_bb_chuva <- adipart(y = m_cont_bb.chuva, x = bb.chuva, index = "richness", relative = TRUE)
51
52 p_pt_seco <- adipart(y = m_cont_s, x = pt.seco, index = "richness", relative = TRUE)
53 p_pt_chuva <- adipart(y = m_cont_c, x = pt.chuva, index = "richness", relative = TRUE)
54
55 # -----
56 # 3. CONSOLIDAÇÃO DA TABELA 1
57 # -----
58 tabela_1_raw <- rbind(
59   extrair_linha_particao(p_ca_seco, "Riacho Caiana (CA)", "Seco"),
60   extrair_linha_particao(p_ca_chuva, "Riacho Caiana (CA)", "Chuvoso"),
61   extrair_linha_particao(p_bb_seco, "Riacho Barro Branco (BB)", "Seco"),
62   extrair_linha_particao(p_bb_chuva, "Riacho Barro Branco (BB)", "Chuvoso"),
63   extrair_linha_particao(p_pt_seco, "Partição Total (PT)", "Seco"),
64   extrair_linha_particao(p_pt_chuva, "Partição Total (PT)", "Chuvoso")
65 )
66
67 # -----

```

```

68 # 4. IMPRESSÃO DA TABELA GT
69 # -----
70 tabela_1_final <- tabela_1_raw %>%
71   gt(groupname_col = "Sistema") %>%
72   tab_header(
73     title = md("**Tabela 1.** Partição aditiva da riqueza de espécies zooplanctônicas nas di
74   ) %>%
75   cols_label(
76     Período      = "Período",
77     Componente   = "Componente Hierárquico",
78     Observado    = "Observado (%)",
79     Esperado     = "Esperado (%)",
80     SES          = "SES",
81     p_valor      = "p-valor"
82   ) %>%
83   cols_align(
84     align = "center",
85     columns = c("Período", "Observado", "Esperado", "SES", "p_valor")
86   )
87 tabela_1_final
88
89 # Formatar nomes das colunas diretamente
90 tabela_1_simples <- tabela_1_raw
91 colnames(tabela_1_simples) <- c("Sistema", "Período", "Componente", "Obs (%)", "Esp (%)", "SI
92
93 # Imprimir no console
94 print(tabela_1_simples, row.names = FALSE)
95
96 # Exportando dados para Excel
97 library(openxlsx)
98 #write.xlsx(tabela_1_simples, file = "res/tabela de partição.xlsx", rowNames = FALSE)
99 wb <- loadWorkbook("res/tabela de partição.xlsx")
100 writeData(wb, sheet = "Sheet 1", x = tabela_1_simples)
101 saveWorkbook(wb, "res/tabela de partição.xlsx", overwrite = TRUE)
102
103 # Tabela final ajustada no Excel
104 library(readxl)
105 part_gttable <- read_excel(
106   path = "res/tabela de partição.xlsx",
107   sheet = "tabela_final",
108   range = "A1:G35") #ajustar para o tamanho da tabela final
109 part_gttable
110 library(gt)

```

```

111 part_gtttable <- gt(part_gtttable)
112 part_gtttable <- sub_missing(part_gtttable,
113                               columns = everything(), missing_text = "")
114 part_gtttable
115 gtsave(part_gtttable, "res/gt-part_gtttable_xlsx.html")
116 saveRDS(part_gtttable, "res/gt-part_gtttable_xlsx.rds")

```

4 Discussion

Depth and width showed greater variations across sites as a result from the fact the streams are barred at some points from roads for local traffic and small dams for water storage. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs.

Conclusions

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Ellen Gomes da Silva: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Investigation.

Ludmilla Cavalcanti Antunes Lucena: Supervision, Validation.

Thais Xavier de Melo: Supervision, Validation.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Ethical approval

Activities inside the REBIO Guaribas were authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license no. SISBIO-89415-1, SISBIO-89429-2 and SISBIO-89415-2).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. No AI was used for data analysis, interpretation, or generation of scientific conclusions. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq 2026).

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Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W).

Figure 2: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by stream. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 3: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by season. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 4: Individual-based extrapolation and rarefaction curves for the richness of resting stages of zooplanktonic taxa in each study stream. Shaded areas indicate 95% confidence intervals.

Figure 5: Classification analysis (UPGMA using Bray-Curtis dissimilarity) of wetland macroinvertebrate samples obtained in a. permanent and b. intermitent wetlands. The first number of the sample label indicates the wetland. The second number indicates the habitat type (1=hydrophytes; 2=reed-like emergent plants; and 3=leafy emergent plants). The third number indicates the individual sweeps realized in each habitat (1–3)UPGMA Hierarchical flexible beta clustering ($\beta = -0.25$) based on Bray-Curtis distance of (a) macroinvertebrates and (b) Chironomidae in the Paraíba River during the 2015 wet season. Tag codes indicate different sections of the river: U = upper segment, M = middle segment, and T = Taperoá River segment. R = reaches (1–3), S = sites (1–3), and Hab = habitat types (1–3). Percent chaining <4%.

Figure 6: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1: subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Table 1: Environmental variables averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 2: Nutrient concentration (g L^{-1}) averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 3: Density (ind/m2) and frequency of occurrence (FO%) of zooplankton collected during the 2024 hydrological cycle.

Table 4: Contribution (expressed in percentage values) of each spatial scale assessed to the total diversity (richness) of the dormant stages of the zooplankton. Numbers in ‘P value’ column indicate the proportion of randomized samples containing more species than the observed sample for each partition.

Figures

Figure 1

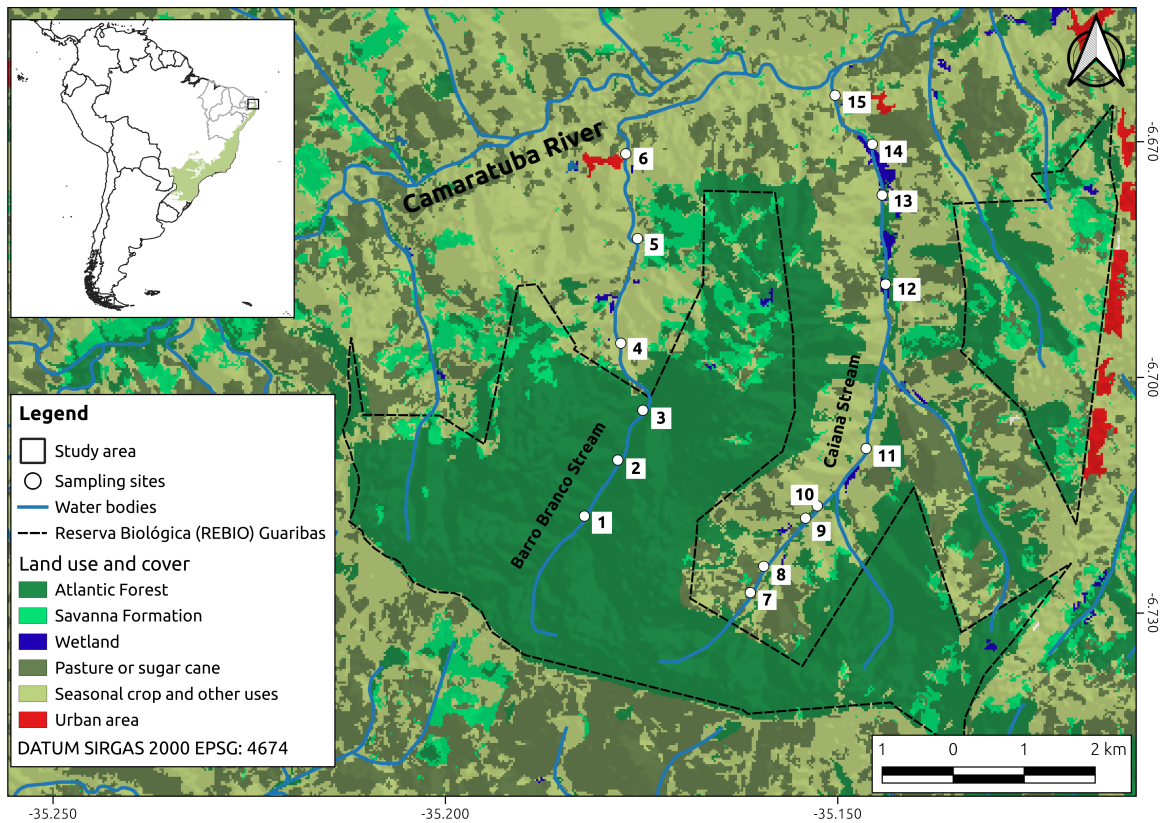


Figure 1: Study area

Figure 2

Figure 3

Figure 4

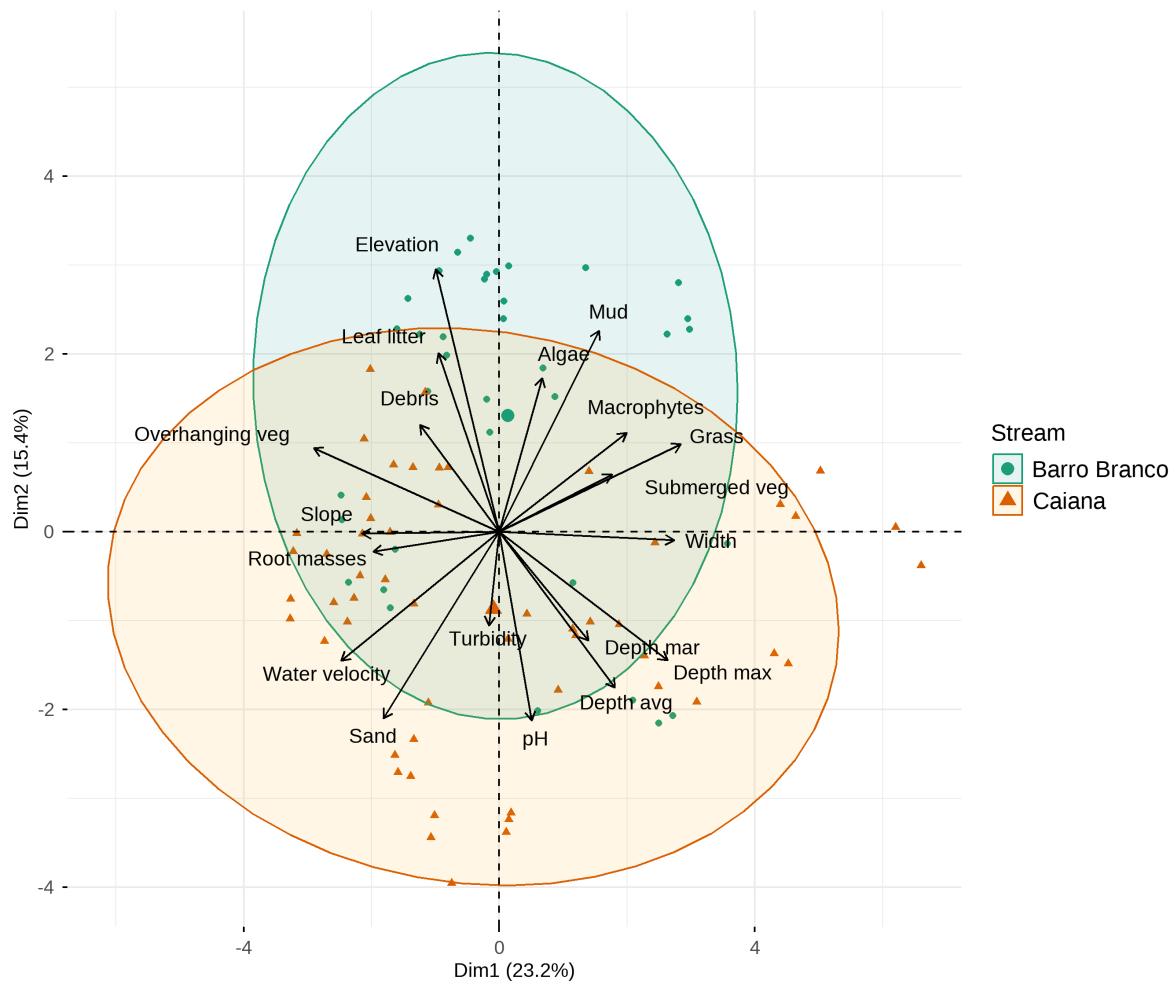


Figure 2: Principal Component Analysis, streams

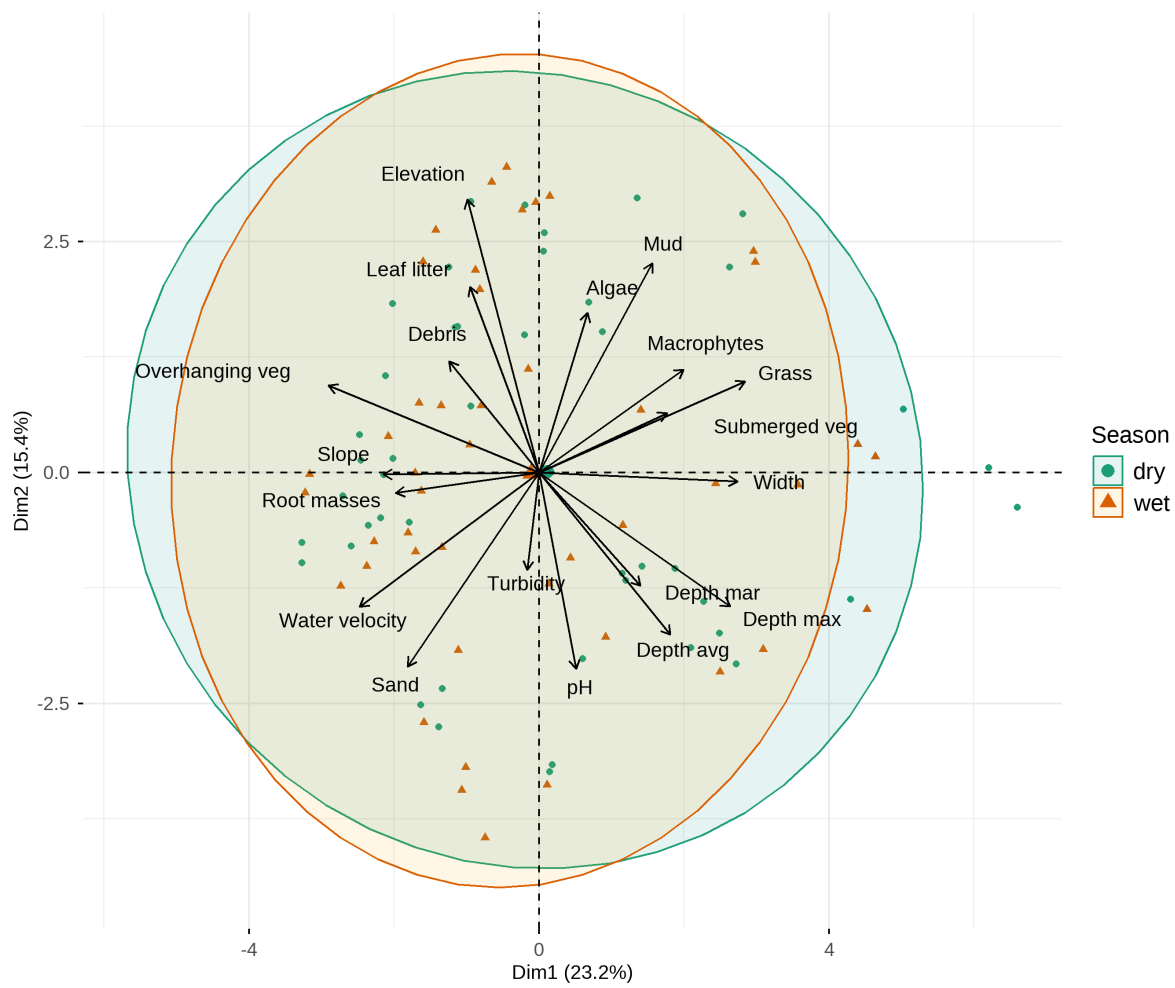


Figure 3: Principal Component Analysis, seasons

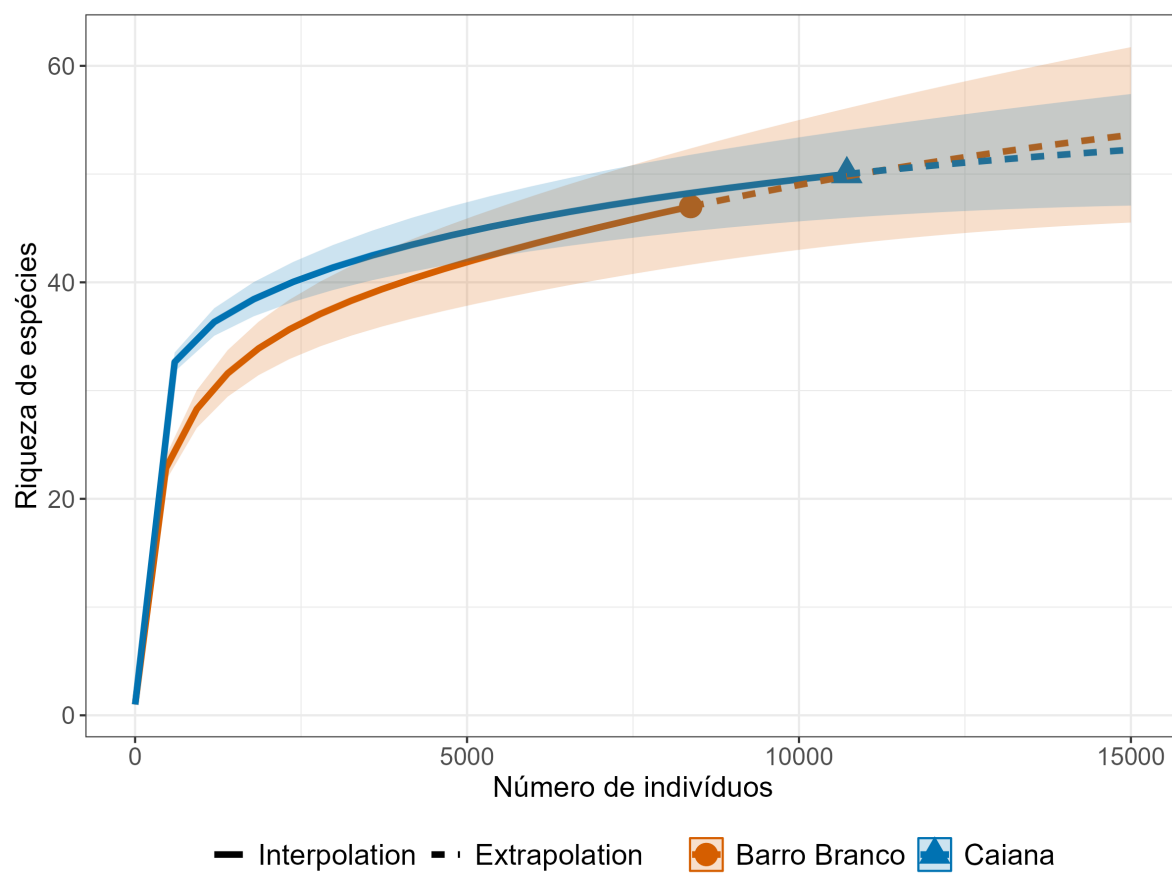


Figure 4: Accumulation curves

Figure 5

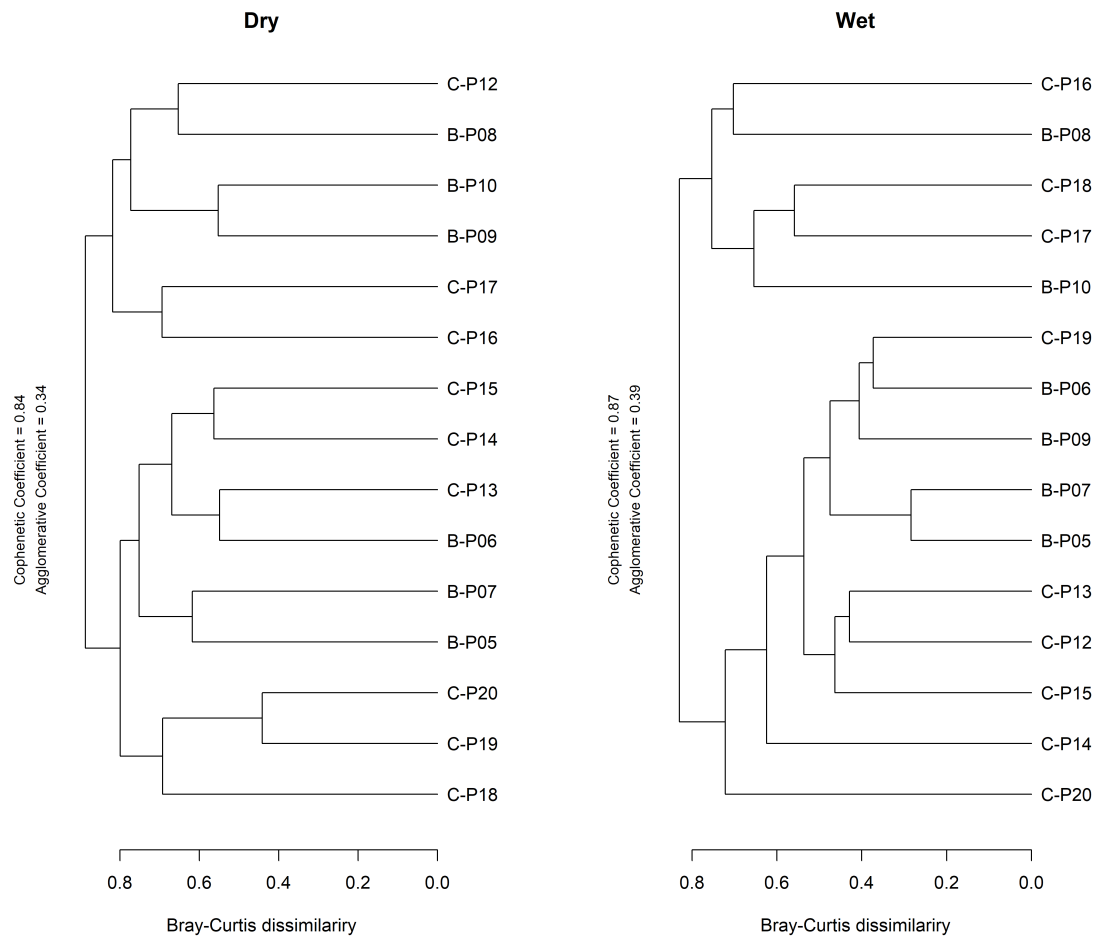


Figure 5: UPGMA Classification

Figure 6

Tables

- Table 1
- Table 2
- Table 3
- Table 4

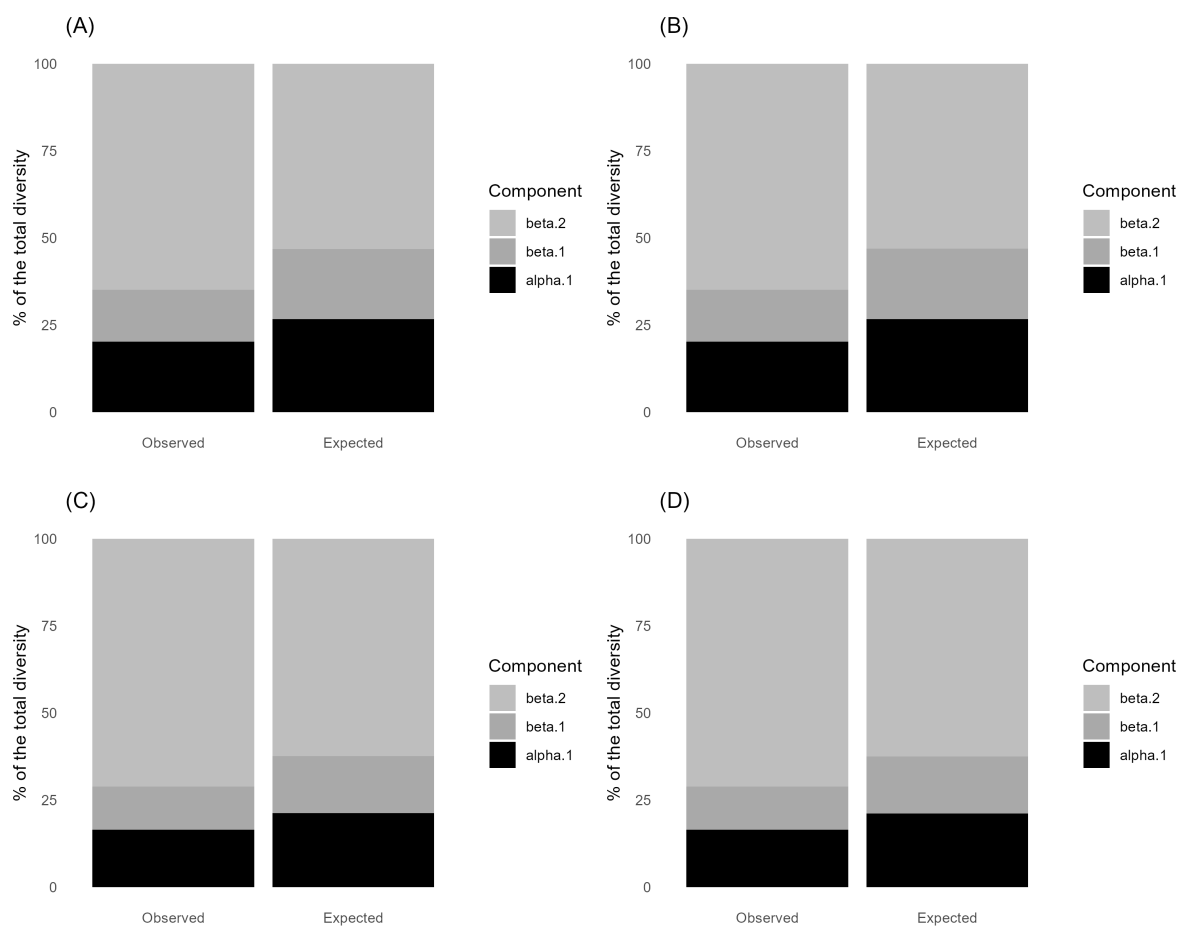


Figure 6: Partitioning plots

Table 1: Environment data GT table.

...1	Barro Branco stream	...3	...4	...5
	Site 1	Site 2	Site 3	Site 4
Water quality				
Temperature (°C)*	25.8 (24.9-27.3)	26.5 (25.3-27.5)	25.2 (24.9-25.5)	27.8 (27.3-28.3)
Dissolved oxygen (mg/L)*	8.1 (7.8-8.5)	7.5 (7.2-7.8)	7.9 (7.7-8)	7.5 (6.5-8.5)
Water velocity (m/s)	0.073 (0.025-0.161)	0.086 (0.035-0.24)	0.135 (0.022-0.207)	0.036 (0.01-0.1)
Conductivity (µS/cm)	137.5 (119.2-164.4)	148 (134.1-156.8)	137.5 (98.3-160.2)	154.2 (141.5-166.9)
Turbidity (NTU)	5 (2-10.7)	4.9 (2.1-14.1)	4.9 (1.5-9.2)	5.7 (4-6.4)
pH	5.9 (4.7-7.5)	5.7 (4.6-7.7)	5.1 (4.5-6)	5.5 (5-6)
Morphology				
Elevation (m a.s.l.)	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (89.1-89.1)
Slope	1.4 (1-2)	2.7 (2.5-3)	2.4 (1.5-3)	1.5 (1-2)
Average depth (cm)	12.8 (9-14)	23.7 (16.3-31.7)	23.8 (16-41)	43.3 (28-58)
Margin depth (cm)	11.8 (3-16)	21.7 (8-30)	18.8 (7-42)	48.3 (16-80)
Maximum depth (cm)	15.5 (13-17)	32.7 (27-40)	29.3 (20-47)	50.8 (35-66)
Width (m)	6.2 (3.6-14.1)	4.3 (0.7-9.2)	2 (1.1-3.2)	7.6 (2-16)
Substrate composition (%)				
Mud	100 (100-100)	100 (100-100)	95 (90-100)	74.2 (5-100)
Sand	0	0	5 (0-10)	25.8 (0-95)
Gravel	0	0	0	0
Rocks	0	0	0	0
Habitat structures (%)				
Aquatic macrophytes	5.8 (5-10)	10 (0-30)	0 (0-0)	17.9 (0-60)
Algae	2.5 (0-5)	0.2 (0-1)	5.5 (0-21)	19.2 (15-25)
Submerged vegetation	7.5 (5-15)	8.3 (5-15)	5.8 (0-10)	5.8 (0-10)
Overhanging vegetation	68.3 (40-90)	70.8 (50-85)	76.7 (60-90)	31.7 (25-40)
Littoral grass	3.3 (0-10)	8.3 (5-15)	6 (0-15)	27.1 (15-40)
Leaf litter	7.5 (5-10)	11 (1-25)	15.8 (5-30)	18.3 (10-30)
Root masses	5 (0-15)	10.8 (5-15)	12.5 (5-20)	4.2 (0-10)
Wood debris	30.8 (20-45)	12.7 (5-20)	17.7 (5-30)	21.7 (5-30)

Table 2: Nutrient concentrations at the sampling sites.

...1	Barro Branco stream	...3	...4	...5	.
	Site 1	Site 2	Site 3	Site 4	S
Ammonia ($\mu\text{g L}^{-1}$)	54.8 (39.3-64.3)	58.9 (47.5-77)	72 (33.4-115.7)	81.6 (68.9-93)	8
Nitrite ($\mu\text{g L}^{-1}$)	0	0	0	0	0
Nitrate ($\mu\text{g L}^{-1}$)	6.0 (0-9.7)	8.8 (6.9-12.6)	5.5 (0-12.6)	18.8 (11.1-31.1)	1
Orthophosphate ($\mu\text{g L}^{-1}$)	0	0	1.0 (0-3)	0	0
Phosphorus ($\mu\text{g L}^{-1}$)	0	4.6 (0.3-10.3)	50.1 (0-150.3)	9 (0-27)	1

Apendices

Non-used figures and tables

 Non-used codes

1 `table(1:10)`

Table 3: Density (ind/m²) taxa GT table.

Stream	Barro Branco Stream	...3	...4	...5	...6	...7	...8
Season	Wet						Dr
Site	Ponto5	Ponto6	Ponto7	Ponto8	Ponto9	Ponto10	Po
Alo.dadayi	0 (0)	0 (0)	0 (0)	5.3 (6.6)	0 (0)	0.4 (0.3)	0 (0)
Alo.hamulata	0 (0)	0 (0)	0 (0)	1.1 (1.3)	0 (0)	0.1 (0.1)	0 (0)
Alonella.sp	0 (0)	0 (0)	0 (0)	0.4 (0.5)	0 (0)	0 (0.1)	0 (0)
Bdelloidea	1.7 (0.5)	1.3 (0)	1.5 (0.5)	3.4 (0.8)	2 (0.8)	2.9 (1.3)	0.6
Bra.patulus	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Cephalodella.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)
Colurella.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Copepodite	0.5 (0.4)	0.1 (0)	0.1 (0)	12.1 (14.5)	0 (0)	0.5 (0.8)	0 (0)
Chydorus.sp	0 (0)	0 (0)	0 (0)	3.2 (3.5)	0 (0)	0.2 (0.1)	0 (0)
Dicranophoroides.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Dia.birgei	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)	0.1 (0.1)	0 (0)
Dia.spinulosum	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Dis.aculeata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Euc.dilatata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Fil.longiseta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Ily.spinifer	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)
Lec.aculeata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.bulla	0 (0)	0 (0)	0 (0)	0 (0.1)	0 (0)	0 (0)	0 (0)
Lec.cornuta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.crepida	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.curvicornis	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.furcata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.leontina	0 (0)	0 (0)	0 (0)	0.5 (0.5)	0 (0)	0 (0.1)	0 (0)
Lec.lunaris	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.luna	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.monostyla	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp2	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.papuana	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.quadridentata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp3	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Leydigia.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lepadella.sp1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lepadella.sp2	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lep.ovalis	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lep.patella	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mesocyclops.sp	0 (0)	0 (0)	0 (0)	0.6 (0.6)	0 (0)	0 (0)	0 (0)
Macrocylops.sp	0 (0)	0 (0)	0 (0)	0.2 (0.2)	0 (0)	0 (0)	0 (0)
Macrochaetus.sp	0 (0)	0 (0)	0 (0)	0.2 (0.3)	0 (0)	0 (0)	0 (0)
Macrothrix.sp	0 (0)	80 0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mac.mira	0 (0)	0 (0)	0 (0)	0.2 (0.3)	0 (0)	0 (0)	0 (0)
Mac.laticornis	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)	0 (0)	0 (0)
Moi.minuta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Myt.crassipes	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mytilina.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Myt.bisulcata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
N.elli	0.2 (0.2)	0.1 (0.1)	0.1 (0)	2.2 (2.6)	0 (0.1)	0.2 (1.2)	0 (0)

Table 4: Partition data GT table.

Sistema	Período	Componente	Obs (%)	Esp (%)	SES	p-valor
Riacho Caiana (CA)	Seco	1 (micro-habitat)	16.40%	21.17%	-12.27	< 0.00
Riacho Caiana (CA)	Seco	2 (ponto)	28.84%	37.52%	-9.32	< 0.00
Riacho Caiana (CA)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Caiana (CA)	Seco	1 (entre micro-habitats)	12.43%	16.35%	-4.84	< 0.00
Riacho Caiana (CA)	Seco	2 (entre pontos)	71.16%	62.48%	9.32	< 0.00
Riacho Caiana (CA)	Chuvoso	1 (micro-habitat)	16.40%	21.18%	-12.12	< 0.00
Riacho Caiana (CA)	Chuvoso	2 (ponto)	28.84%	37.52%	-9.29	< 0.00
Riacho Caiana (CA)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Caiana (CA)	Chuvoso	1 (entre micro-habitats)	12.43%	16.33%	-4.70	< 0.00
Riacho Caiana (CA)	Chuvoso	2 (entre pontos)	71.16%	62.48%	9.29	< 0.00
Riacho Barro Branco (BB)	Seco	1 (micro-habitat)	20.23%	26.63%	-11.64	< 0.00
Riacho Barro Branco (BB)	Seco	2 (ponto)	35.04%	46.86%	-9.03	< 0.00
Riacho Barro Branco (BB)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Barro Branco (BB)	Seco	1 (entre micro-habitats)	14.81%	20.23%	-4.66	< 0.00
Riacho Barro Branco (BB)	Seco	2 (entre pontos)	64.96%	53.14%	9.03	< 0.00
Riacho Barro Branco (BB)	Chuvoso	1 (micro-habitat)	20.23%	26.64%	-11.86	< 0.00
Riacho Barro Branco (BB)	Chuvoso	2 (ponto)	35.04%	46.88%	-9.39	< 0.00
Riacho Barro Branco (BB)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Barro Branco (BB)	Chuvoso	1 (entre micro-habitats)	14.81%	20.24%	-4.88	< 0.00
Riacho Barro Branco (BB)	Chuvoso	2 (entre pontos)	64.96%	53.12%	9.39	< 0.00
Partição Total (PT)	Seco	1 (micro-habitat)	12.45%	19.41%	-21.58	< 0.00
Partição Total (PT)	Seco	2 (ponto)	22.18%	34.41%	-16.50	< 0.00
Partição Total (PT)	Seco	3 (riacho)	73.28%	84.12%	-5.60	< 0.00
Partição Total (PT)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Partição Total (PT)	Seco	1 (entre micro-habitats)	9.73%	15.00%	-8.68	< 0.00
Partição Total (PT)	Seco	2 (entre pontos)	51.09%	49.72%	0.69	0.499
Partição Total (PT)	Seco	3 (entre riachos)	26.72%	15.88%	5.60	< 0.00
Partição Total (PT)	Chuvoso	1 (micro-habitat)	23.72%	51.72%	-45.73	< 0.00
Partição Total (PT)	Chuvoso	2 (ponto)	35.50%	69.45%	-35.41	< 0.00
Partição Total (PT)	Chuvoso	3 (riacho)	83.75%	96.48%	-9.42	< 0.00
Partição Total (PT)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Partição Total (PT)	Chuvoso	1 (entre micro-habitats)	11.78%	17.73%	-7.46	< 0.00
Partição Total (PT)	Chuvoso	2 (entre pontos)	48.25%	27.04%	13.64	< 0.00
Partição Total (PT)	Chuvoso	3 (entre riachos)	16.25%	3.52%	9.42	< 0.00